MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE BDD

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/117-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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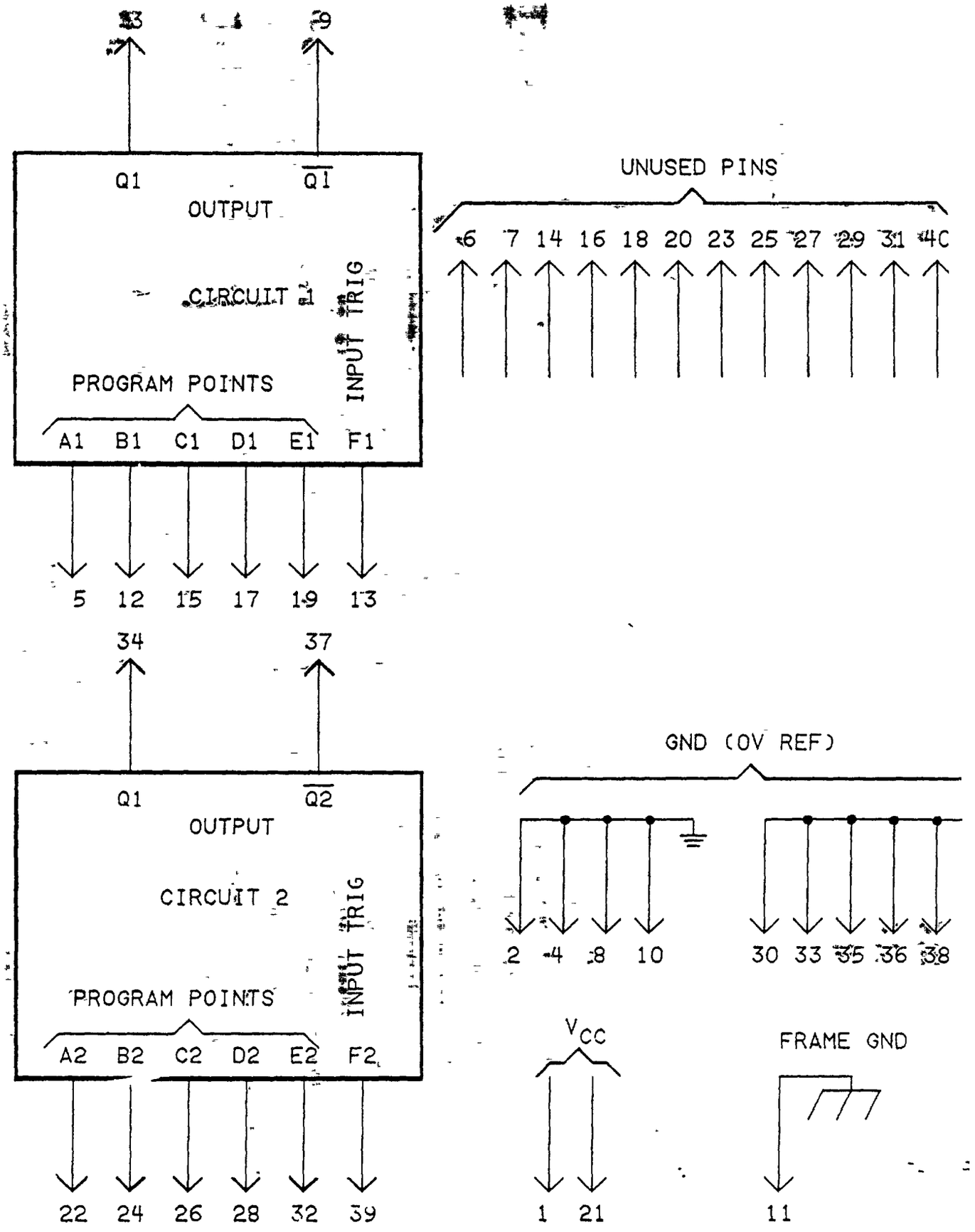


FIGURE 1. BDD module function diagram, monostable multivibrator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A7	Continuity	II	X	0	1	0	1	0	1	0	1	Ω
A8	Continuity	II	S	0	1	0	1	0	1	0	1	Ω
B1,B2	Isolation	II	X	10	∞	10	∞	10	∞	10	∞	MΩ
C1,C4,C5,C8	Output voltage, low level	III	X	0.0	0.4	0.00	0.45	0.0	0.4	0.00	0.45	V
C2,C3,C6,C7	Output voltage, high level	III	X	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
D1-D4	Output short circuit current	IV	X	10	65	10	65	10	65	10	65	mA
D5,D7	Input current, high level	IV	X	0	4	0	8	0	6	0	10	μA
D6,D8	Input current, low level	IV	X	0.0	3.2	0.0	3.2	0.0	3.2	0.0	3.	mA
E1	Power supply current, quiescent	V	S	0	120	0	120	0	120	0	12	mA
E2	Power supply current, operating	V	S	0	120	0	120	0	120	0	120	mA
F1-F4	Output pulse width, 0.1 μs	VI	X	0.090	0.120	0.088	0.122	0.087	0.123	0.085	0.125	μs
F5-F8	Output pulse width, 0.2 μs	VI	X	0.180	0.240	0.176	0.244	0.174	0.246	0.170	0.250	μs
F9-F12	Output pulse width, 0.3 μs	VI	S	0.255	0.345	0.250	0.350	0.245	0.355	0.240	0.360	μs
F13-F16	Output pulse width, 0.4 μs	VI	X	0.350	0.440	0.343	0.447	0.377	0.453	0.330	0.460	μs
F17-F20	Output pulse width, 0.5 μs	VI	S	0.465	0.535	0.455	0.545	0.450	0.550	0.440	0.560	μs
F21-F24	Output pulse width, 0.6 μs	VI	S	0.558	0.642	0.546	0.654	0.540	0.660	0.528	0.672	μs
F25-F28	Output pulse width, 0.7 μs	VI	S	0.651	0.749	0.637	0.763	0.630	0.770	0.616	0.784	μs
F29-F32	Output pulse width, 0.8 μs	VI	X	0.744	0.856	0.728	0.872	0.720	0.880	0.704	0.896	μs
F33-F36	Output pulse width, 0.9 μs	VI	S	0.837	0.963	0.819	0.981	0.810	0.990	0.792	1.0	μs
F37-F40	Output pulse width, 1.0 μs	VI	S	0.930	1.070	0.910	1.090	0.90	1.10	0.88	1.12	μs
F41-F44	Output pulse width, 1.1 μs	VI	S	1.02	1.18	1.00	1.20	0.99	1.21	0.97	1.23	μs
F45-F48	Output pulse width 1.2 μs	VI	S	1.12	1.28	1.09	1.31	1.08	1.32	1.06	1.34	μs
F49-F52	Output pulse width 1.3 μs	VI	S	1.21	1.39	1.18	1.42	1.17	1.43	1.14	1.46	μs
F53-F56	Output pulse width, 1.4 μs	VI	S	1.30	1.50	1.27	1.53	1.26	1.54	1.23	1.57	μs
F57-F60	Output pulse width, 1.5 μs	VI	X	1.39	1.61	1.36	1.64	1.35	1.65	1.32	1.68	μs

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G1-G4	Transition time (t_{THL})	VII	S	6	40	6	45	6	40	6	45	ns
G5-G8	Transition time (t_{TLH})	VII	S	8	90	8	90	8	90	8	90	ns
H1,H3	Propagation delay time (t_{PLH})	VIII	S	10	100	10	100	10	100	10	100	ns
H2,H4	Propagation delay time (t_{PHL})	VIII	S	10	100	10	100	10	100	10	100	ns
J1-J4	Output pulse amplitude, low level	IX	S	0.0	0.4	0.00	0.45	0.0	0.4	0.00	0.45	V
J5-J8	Output pulse amplitude, high level	IX	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
K1-K4	Output pulse, overshoot	X	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K5-K8	Output pulse, undershoot	X	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K9-K12	Output pulse, preshoot	X	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
L1,L3	Pulse width recovery time	XI	S	0.09	0.12	0.088	0.122	0.087	0.123	0.085	0.125	µs
L2,L4	Pulse width recovery time	XI	S	1.39	1.61	1.36	1.64	1.35	1.65	1.32	1.68	µs
M1-M4	Input sensitivity, trigger	XII	S	0.09	0.12	0.088	0.122	0.087	0.123	0.085	0.125	µs
M5-M8	Input immunity, 20 ns	XII	S	0	5	0	5	0	5	0	5	ns
M9-M12	Input immunity, 0.95 V	XII	S	0	5	0	5	0	5	0	5	ns
N1,N2	Power supply transient rise time	XIII	S	9.0	53.0	9.0	53.0	9.0	53.0	9.0	53.0	µs
N3,N4	Power supply transient rise time	XIII	S	1.5	40.0	1.5	40.0	1.5	40.0	1.5	40.0	µs
N5-N8	Power supply false trigger	XIII	S	2.0	5.5	2.0	5.5	2.0	5.5	2.0	5.5	V
P1-P12	Crosstalk	XIV	S	0	5	0	5	0	5	0	5	ns
Q1,Q2	Zener applied potential	XV	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
Q3,Q4	Trigger voltage high	XV	S	0.0	0.4	0.00	0.45	0.0	0.4	0.00	0.45	V
R1,R2	V_{OUT} (low-loaded)	XVI	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test condition Q or R is part of a sequence of tests, the order-of-testing shall be as follows: Q, R where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A and C through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A and C through N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests A through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/117 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658637. Part number M28787/117-1 supersedes part numbers 2658637 and M28787/117.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	660 mW at any duty cycle and rated load
Input voltage	
High level	3.2 V minimum, V_{CC} maximum
Low level	0.0 V minimum, 0.95 V maximum
Output pulse width range	0.10 µs minimum, 1.50 µs maximum (in 0.10 µs increments)

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Output drive current maximum capability	
Low level	19.2 mA
High level	200 µA
Input pulse width	85 ns minimum
Recovery time	500 ns minimum

6.5 Module function. The module consists of two monostable multivibrator circu which have output pulse widths programmable from 0.1 µs to 1.50 µs in 0.10 µs increments.

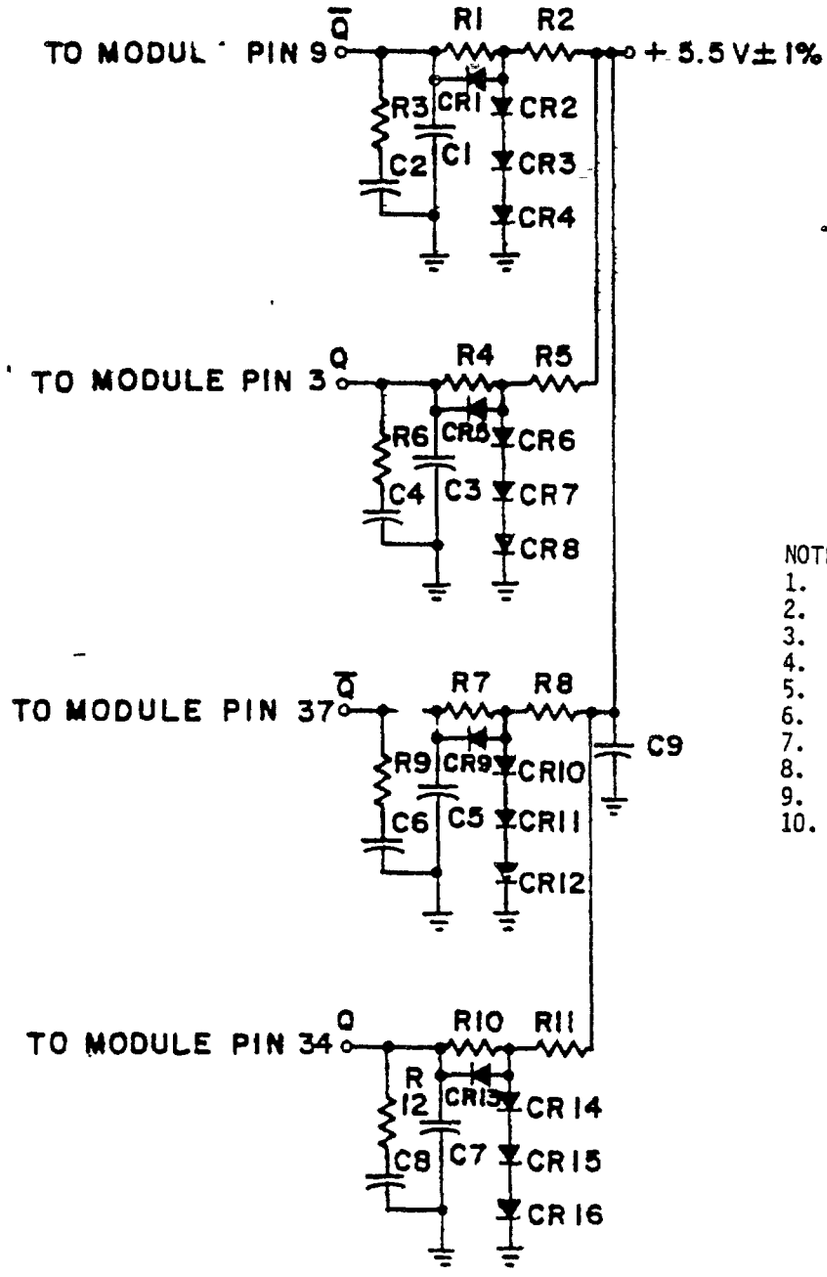
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of t changes.

Agent:
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Preparing activity:
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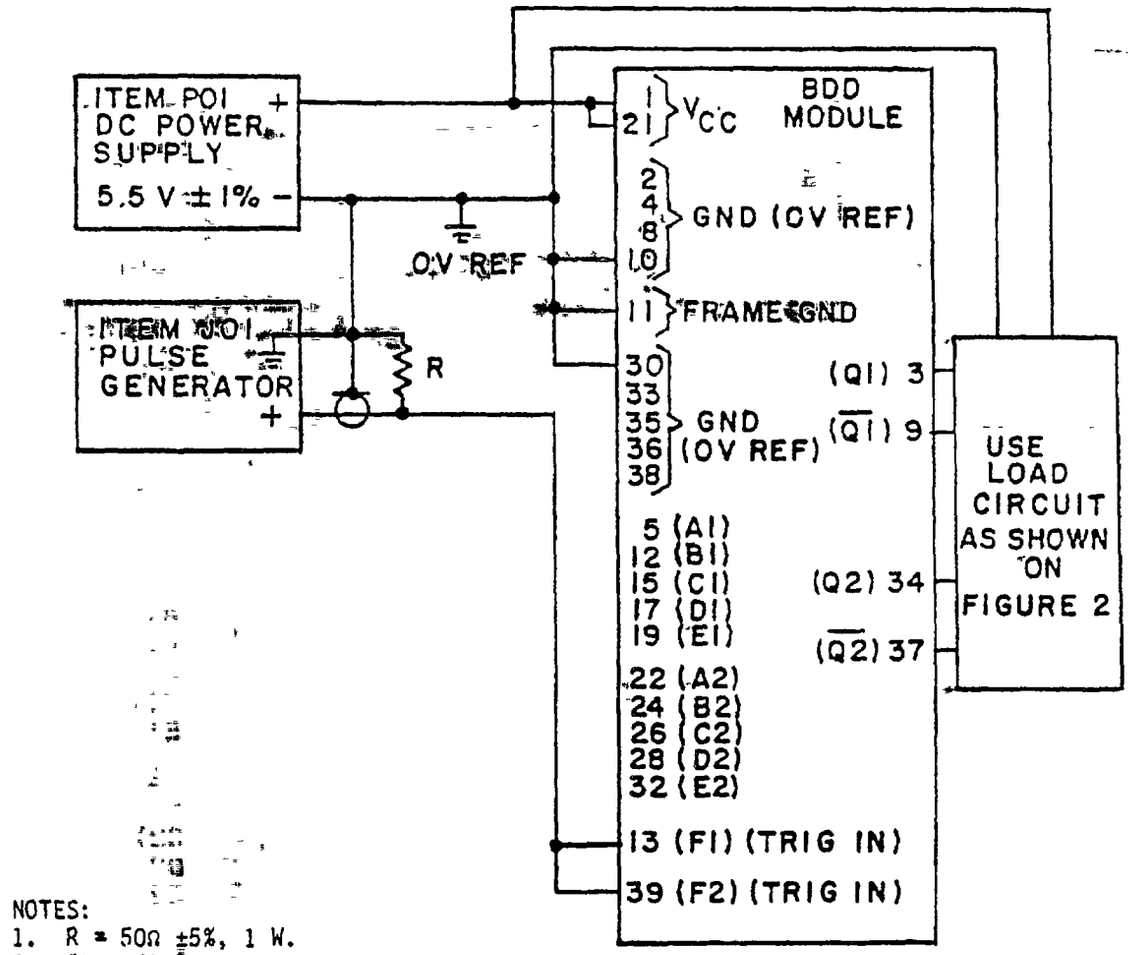
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NOTES:
1. R1= R7 = 8.25kΩ ±1%, 1/8 W.
2. R2= R8 = 604 Ω ±1%, 1/8 W.
3. R3= R6= R9= R12 = 100Ω ±1%, 1/8 W.
4. R4= R10 = 7.5 kΩ ±1%, 1/8 W.
5. R5= R11 = 249 Ω ±1%, 1/8 W.
6. CR1-CR16 = 1N4150 or equivalent.
7. C1= C3= C5= C7 = 33 pF ±10%.
8. C2= C4= C6= C8 = 330 pF ±10%.
9. C9 = 1.0 µF ±10%.
10. Loads shall be assembled on a board less than 3 by 3 (76.2 by 76.2 mm) with components and wiring within 1/2 inch (12.7 mm) of the board.

FIGURE 2. BDD module load circuit.

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NOTES:
1. R = 50Ω ±5%, 1 W.
2. Item J01 setup:
   PA = 5 V ±5%.
   PW = 150 ns ±10%
   (at 1.5 V level).
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 3.0 µs ±10%.

FIGURE 3. Operating temperature, component temperature, and life test circuit.

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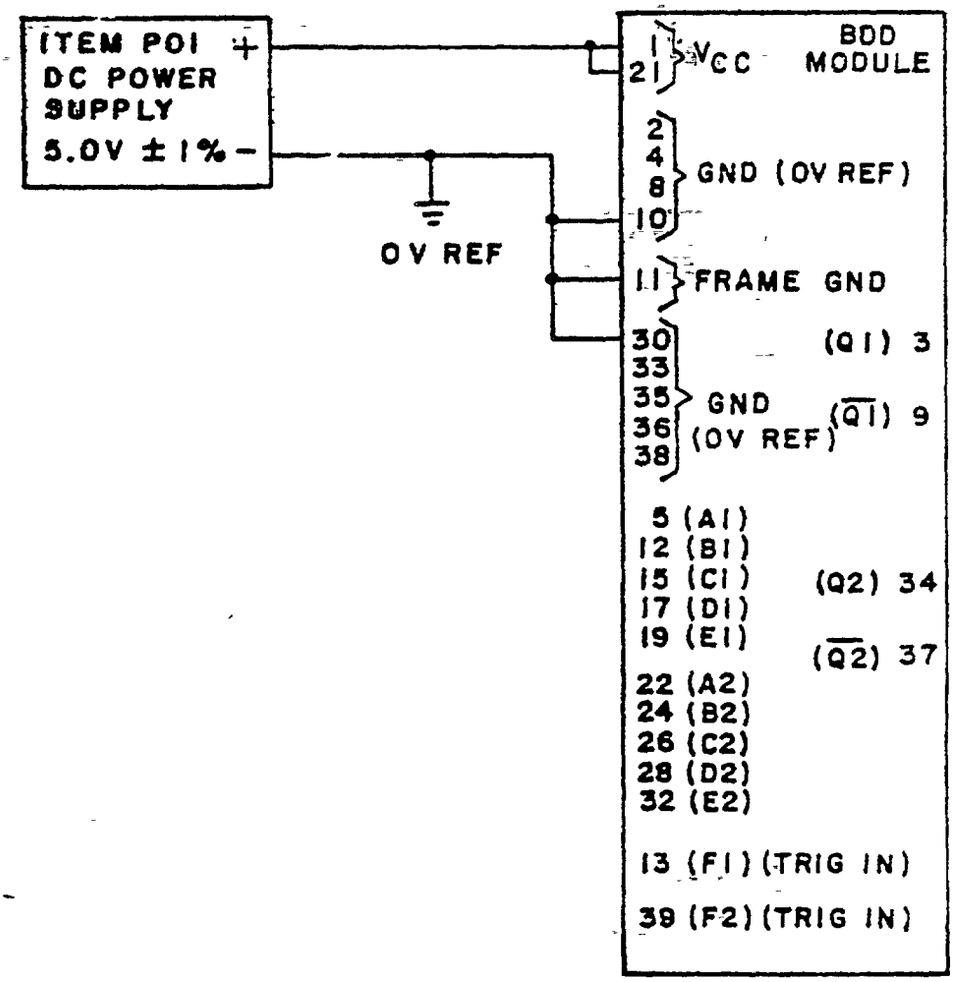


FIGURE 4. Vibration test circuit.

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TABLE II. Module continuity and isolation test.

Test condition	Parameter	Connect item R07 between module pins		Initial limits 25° ±5°C ambient		Units
				Min	Max	
A1	Continuity	10	2	0	1	Ω
A2	↓	10	4	↓	↓	↓
A3		10	8			
A4		30	33			
A5		30	35			
A6		30	36			
A7		30	38			
A8		11	Frame 1/			
B1	Isolation 2/	11	All other pins interconnected	10	∞	MΩ
B2	Isolation 2/	Unused pins 6, 7,14,16,18,20, 23,25,27,29,31, and 40 inter-connected	To all other pins interconnected	10	∞	MΩ

1/ Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead of the ohmmeter.

2/ Test conditions B1 and B2 shall be performed using item R06, megohm bridge, with a test voltage of 100 V dc ±10%.

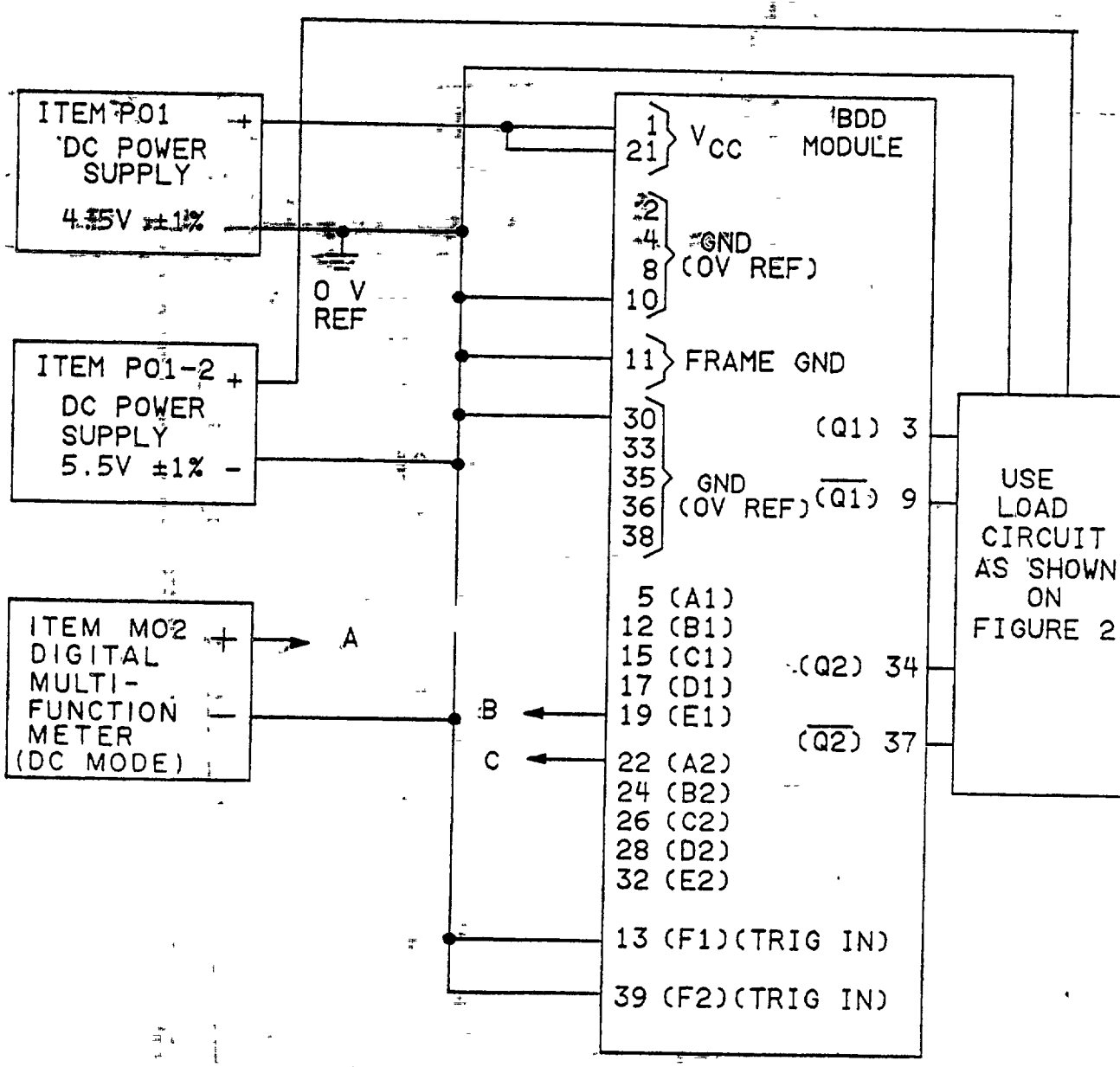
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TABLE III. Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	Q1 Output voltage, low level	3	Open	Open	0.0	0.4	V	5
C2	$\overline{Q1}$ Output voltage, high level	9	Open		3.8	5.5		
C3	Q1 Output voltage, high level	3	10		3.8	5.5		
C4	$\overline{Q1}$ Output voltage, low level	9	10		0.0	0.4		
C5	Q2 Output voltage, low level	34	Open		0.0	0.4		
C6	$\overline{Q2}$ Output voltage, high level	37			3.8	5.5		
C7	Q2 Output voltage, high level	34		30	3.8	5.5		
C8	$\overline{Q2}$ Output voltage, low level	37		30	0.0	0.4		

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NOTES:
1. Prior to making measurements for test conditions C3, C4, C7, and C8, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. Interconnecting conductor length between pins 10 and 19 and between pins 22 and 30 shall not exceed six inches (152.4 mm).

FIGURE 5. Output voltage (low level and high level) test circuit for table III.

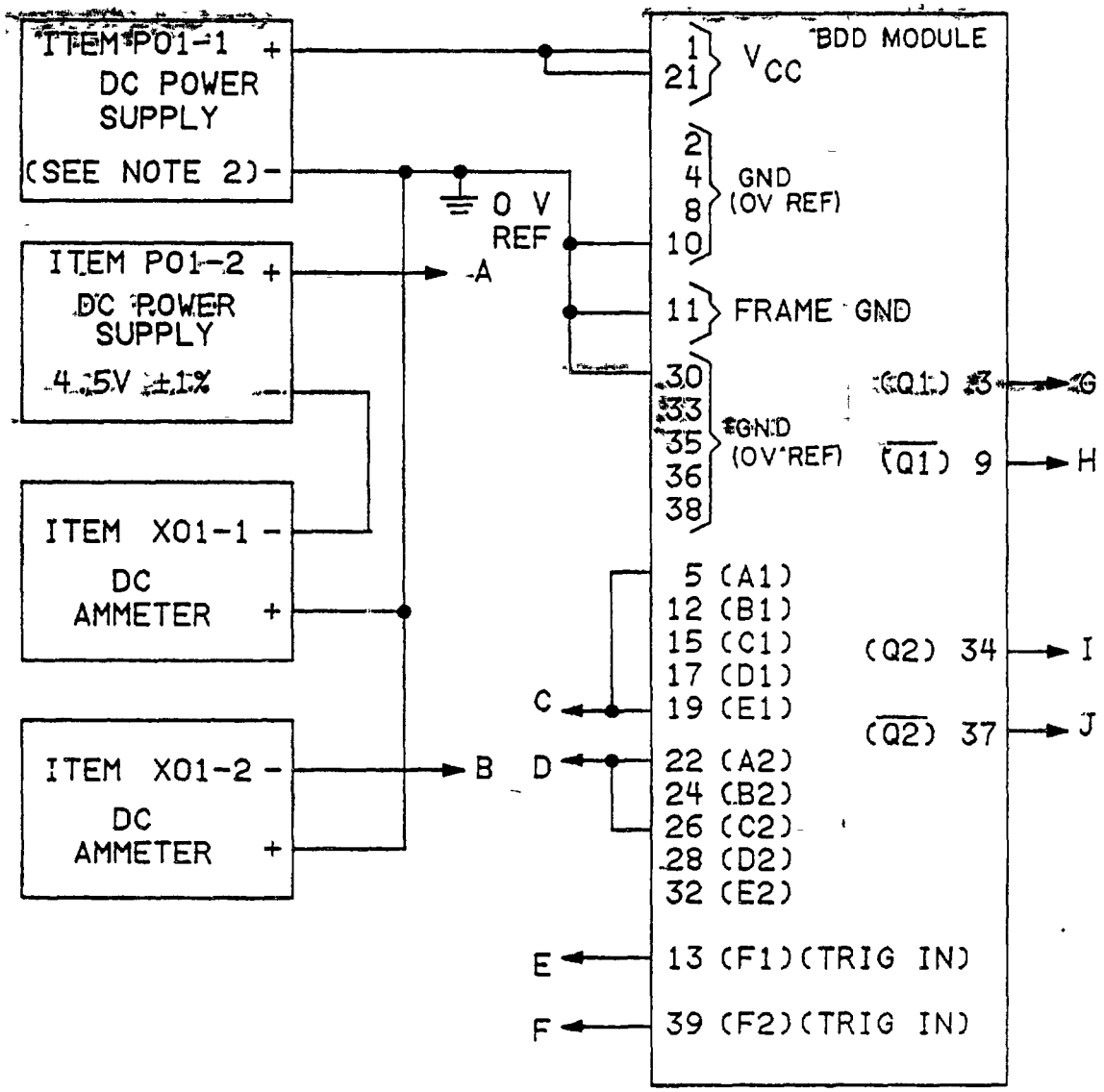
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TABLE IV. Output short circuit current and input current test.

Test condition	Parameter	Connect: C to pin 1/	D to pin 1/	E to pin	F to pin	G to pin	H to pin	I to pin	J to pin	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
D1	Q1 Output short circuit current	10	Open	0V	0V	B	Open	Open	Open	10	65	mA	6
D2	$\overline{Q1}$	Open	Open	0V	0V	Open	B	Open	Open				
D3	Q2		30	0V	0V		Open	B	Open				
D4	$\overline{Q2}$		Open	0V	0V			Open	B				
D5	A1 and C2 Input current, high level			A	Open				Open	0	4	µA	
D6	A1 and C2 Input current, low level			B	Open					0.0	3.2	mA	
D7	A1 and C2 Input current, high level			Open	A					0	4	µA	
D8	A1 and C2 Input current, low level			Open	B					0.0	3.2	mA	

1/ Interconnection length between any two pins specified in these columns shall not exceed six inches (152.4 mm).

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CAUTION
Do not isolate the chassis of ammeter XO1 from earth ground.

CAUTION
The chassis of power supply PO1-2 shall be isolated from earth ground.

NOTES:
1. Prior to making measurements for test conditions D1 and D3, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. Set item PO1-1 to 5.5 V ±1% for test conditions D1-D4, D6, and D8 and 4.5 V ±1% for test condition D5 and D7.

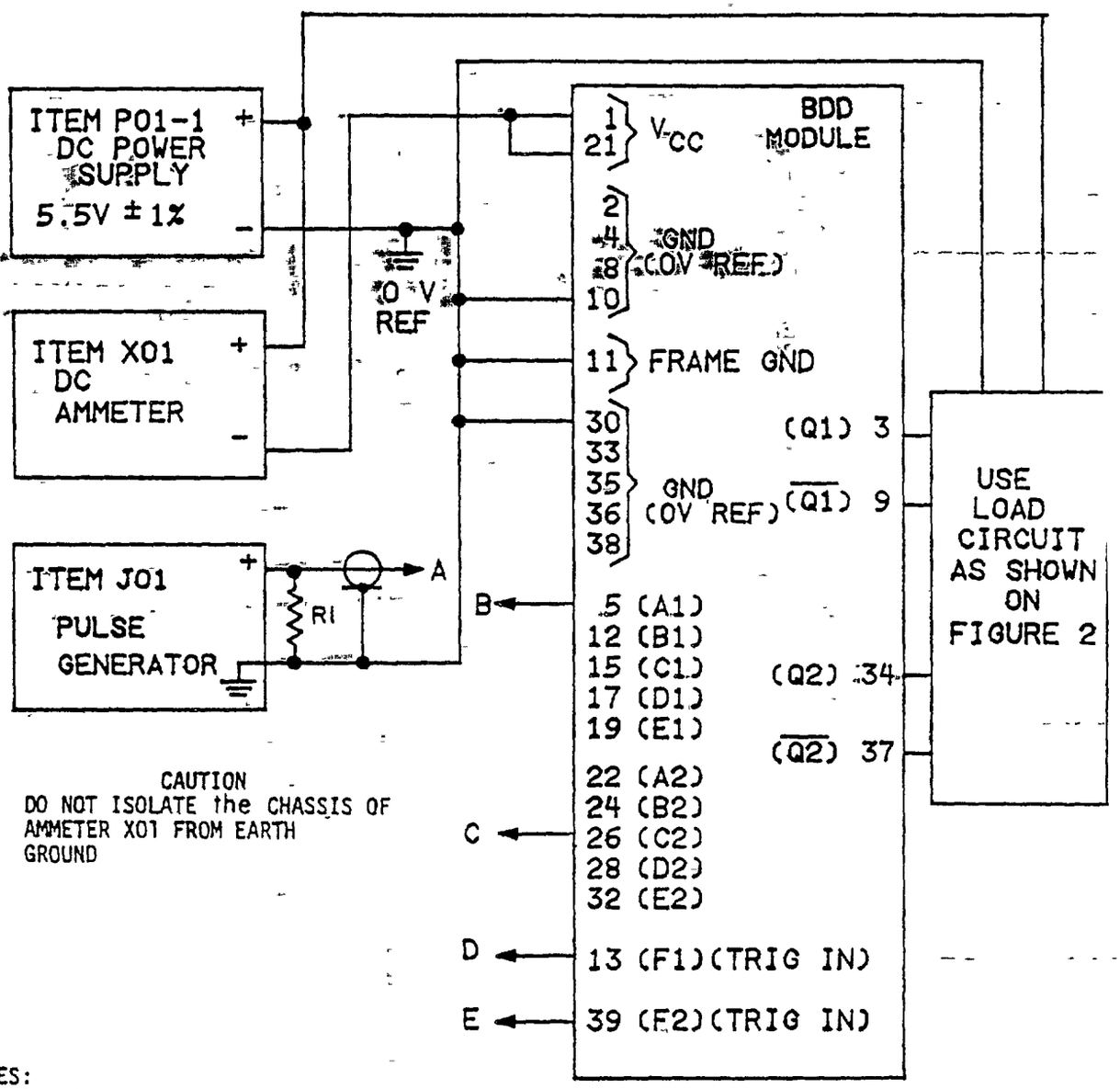
FIGURE 6. Output short circuit current and input current test circuit for table IV.

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TABLE V. Power supply current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figu
		B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	Power supply current, quiescent	19	22	10	10	0	120	mA	7
E2	Power supply current, operating	19	22	A	A	0	120	mA	7

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NOTES:
1. R1 = 50 Ω ±5%, 1 W.
2. Item J01 setup:
   PA = 4.0 V ±5%.
   PW = 125 ns ±10% (at the 1.5-V level).
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 600 ns ±10%.

FIGURE 7. Power supply current test circuit for table V.

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TABLE VI. Output pulse width (Q and $\overline{Q}$) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	Q1 Output pulse width, 0.1 µs	5	19	Open	Open	13	3	0.090	0.120	µs	8,9
F2	$\overline{Q1}$	5	19			13	9	0.090	0.120		8,10
F3	Q2	22	26			39	34	0.090	0.120		8,9
F4	$\overline{Q2}$	22	26			39	37	0.090	0.120		8,10
F5	Q1 Output pulse width, 0.2 µs	5	17	12	19	13	3	0.180	0.240		8,9
F6	$\overline{Q1}$	5	17	12	19	13	9	0.180	0.240		8,10
F7	Q2	26	32	22	28	39	34	0.180	0.240		8,9
F8	$\overline{Q2}$	26	32	22	28	39	37	0.180	0.240		8,10
F9	Q1 Output pulse width, 0.3 µs	12	19	Open	Open	13	3	0.255	0.345		8,9
F10	$\overline{Q1}$	12	19			13	9	0.255	0.345		8,10
F11	Q2	22	28			39	34	0.255	0.345		8,9
F12	$\overline{Q2}$	22	28			39	37	0.255	0.345		8,10
F13	Q1 Output pulse width, 0.4 µs	12	17	15	19	13	3	0.350	0.440		8,9
F14	$\overline{Q1}$	12	17	15	19	13	9	0.350	0.440		8,10
F15	Q2	28	32	22	24	39	34	0.350	0.440		8,9
F16	$\overline{Q2}$	28	32	22	24	39	37	0.350	0.440		8,10
F17	Q1 Output pulse width, 0.5 µs	5	12	15	19	13	3	0.465	0.535		8,9
F18	$\overline{Q1}$	5	12	15	19	13	9	0.465	0.535		8,10
F19	Q2	26	28	22	24	39	34	0.465	0.535		8,9
F20	Q2	26	28	22	24	39	37	0.465	0.535		8,10
F21	Q1 Output pulse width, 0.6 µs	5	17	15	19	13	3	0.558	0.642		8,9

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TABLE VI. Output pulse width (Q and $\overline{Q}$) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure**
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F22	$\overline{Q1}$ Output pulse width, 0.6 µs	5	17	15	19	13	9	0.558	0.642	µs	8,10
F23	Q2 Output pulse width, 0.6 µs	26	32	22	24	39	34	0.558	0.642		8,9
F24	$\overline{Q2}$ Output pulse width, 0.6 µs	26	32	22	24	39	37	0.558	0.642		8,10
F25	Q1 Output pulse width, 0.7 µs	15	19	Open	Open	13	3	0.651	0.749		8,9
F26	$\overline{Q1}$	15	19			13	9	0.651	0.749		8,10
F27	Q2	22	24			39	34	0.651	0.749		8,9
F28	$\overline{Q2}$	22	24			39	37	0.651	0.749		8,10
F29	Q1 Output pulse width, 0.8 µs	15	17			13	3	0.744	0.856		8,9
F30	$\overline{Q1}$	15	17			13	9	0.744	0.856		8,10
F31	Q2	24	32			39	34	0.744	0.856		8,9
F32	$\overline{Q2}$	24	32			39	37	0.744	0.856		8,10
F33	Q1 Output pulse width, 0.9 µs	5	15			13	3	0.837	0.963		8,9
F34	$\overline{Q1}$	5	15			13	9	0.837	0.963		8,10
F35	Q2	24	26			39	34	0.837	0.963		8,9
F36	$\overline{Q2}$	24	26			39	37	0.837	0.963		8,10
F37	Q1 Output pulse width, 1.0 µs	5	17	12	15	13	3	0.930	1.070		8,9
F38	$\overline{Q1}$	5	17	12	15	13	9	0.930	1.070		8,10
F39	Q2	26	32	24	28	39	34	0.930	1.070		8,9
F40	$\overline{Q2}$	26	32	24	28	39	37	0 930	1.070		8,10
F41	Q1 Output pulse width, 1.1 µs	12	15	Open	Open	13	3	1.02	1.18		8,9

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TABLE VI. Output pulse width (Q and $\overline{Q}$) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F42	$\overline{Q1}$ Output pulse width, 1.1 µs	12	15	Open	Open	13	9	1.02	1.18	µs	8,10
F43	Q2 Output pulse width, 1.1 µs	24	28			39	34	1.02	1.18		8,9
F44	$\overline{Q2}$ Output pulse width, 1.1 µs	24	28			39	37	1.02	1.18		8,10
F45	Q1 Output pulse width, 1.2 µs	12	17			13	3	1.12	1.28		8,9
F46	$\overline{Q1}$	12	17			13	9	1.12	1.28		8,10
F47	Q2	28	32			39	34	1.12	1.28		8,9
F48	$\overline{Q2}$	28	32			39	37	1.12	1.28		8,10
F49	Q1 Output pulse width, 1.3 µs	5	12			13	3	1.21	1.39		8,9
F50	$\overline{Q1}$	5	12			13	9	1.21	1.39		8,10
F51	Q2	26	28			39	34	1.21	1.39		8,9
F52	$\overline{Q2}$	26	28			39	37	1.21	1.39		8,10
F53	Q1 Output pulse width, 1.4 µs	5	17			13	3	1.30	1.50		8,9
F54	$\overline{Q1}$	5	17			13	9	1.30	1.50		8,10
F55	Q2	26	32			39	34	1.30	1.50		8,9
F56	$\overline{Q2}$	26	32			39	37	1.30	1.50		8,10
F57	Q1 Output pulse width, 1.5 µs	Open	Open			13	3	1.39	1.61		8,9
F58	$\overline{Q1}$					13	9	1.39	1.61		8,10
F59	Q2					39	34	1.39	1.61		8,9
F60	Q2					39	37	1.39	1.61		8,10

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TABLE VII. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
G1	Q1 Transition time (t_{THL})	13	3	6	40	ns	8,9
G2	$\overline{Q1}$	13	9				8,10
G3	Q2	39	34				8,9
G4	$\overline{Q2}$	39	37				8,10
G5	Q1 Transition time (t_{TLH})	13	3	8	90		8,9
G6	$\overline{Q1}$	13	9				8,10
G7	Q2	39	34				8,9
G8	$\overline{Q2}$	39	37				8,10

TABLE VIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
H1	Q1 Propagation delay time (t_{PLH})	13	3	10	100	ns	8,9
H2	$\overline{Q1}$ Propagation delay time (t_{PHL})	13	9				8,10
H3	Q2 Propagation delay time (t_{PLH})	39	34				8,9
H4	$\overline{Q2}$ Propagation delay time (t_{PHL})	39	37				8,10

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TABLE IX. Output pulse amplitude (low-level and high-level) test.

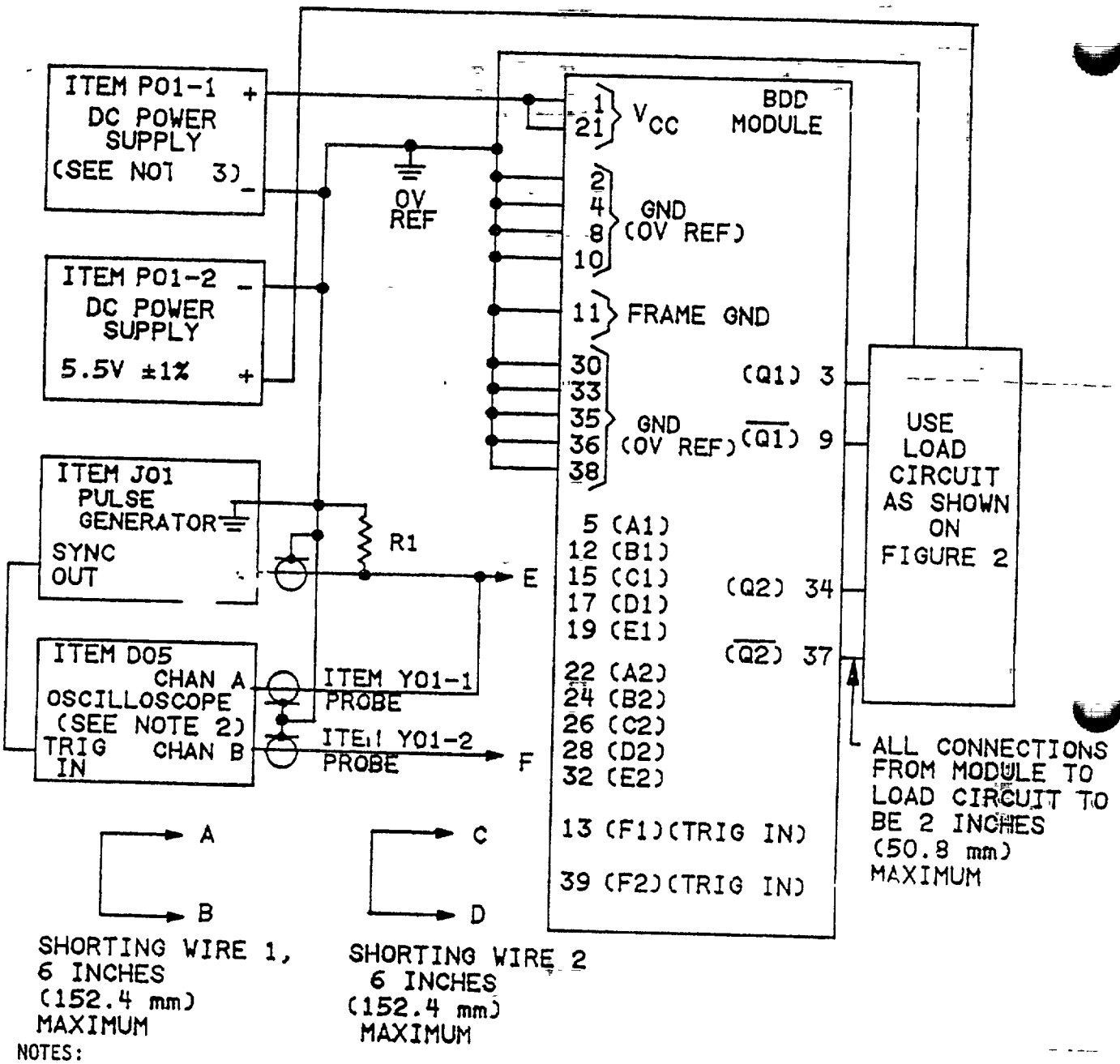
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
J1	Q1 Output pulse amplitude, low level	13	3	0.0	0.4	V	3,9
J2	$\overline{Q1}$	13	9				8,10
J3	Q2	39	34				8,9
J4	$\overline{Q2}$	39	37				8,10
J5	Q1 Output pulse amplitude, high level	13	3	3.8	5.5		8,9
J6	$\overline{Q1}$	13	9				8,10
J7	Q2	39	34				8,9
J8	$\overline{Q2}$	39	37				8,10

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TABLE X. Output pulse (overshoot, undershoot and preshoot) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
K1	Q1 Output pulse, overshoot	13	3	0.0	0.5	V	8,9
K2	$\overline{Q1}$	13	9				8,10
K3	Q2	39	34				8,9
K4	$\overline{Q2}$	39	37				8,10
K5	Q1 Output pulse, undershoot	13	3				8,9
K6	$\overline{Q1}$	13	9				8,10
K7	Q2	39	34				8,9
K8	$\overline{Q2}$	39	37				8,10
K9	Q1 Output pulse, preshoot	13	3				8,9
K10	$\overline{Q1}$	13	9				8,10
K11	Q2	39	34				8,9
K12	$\overline{Q2}$	39	37				8,10

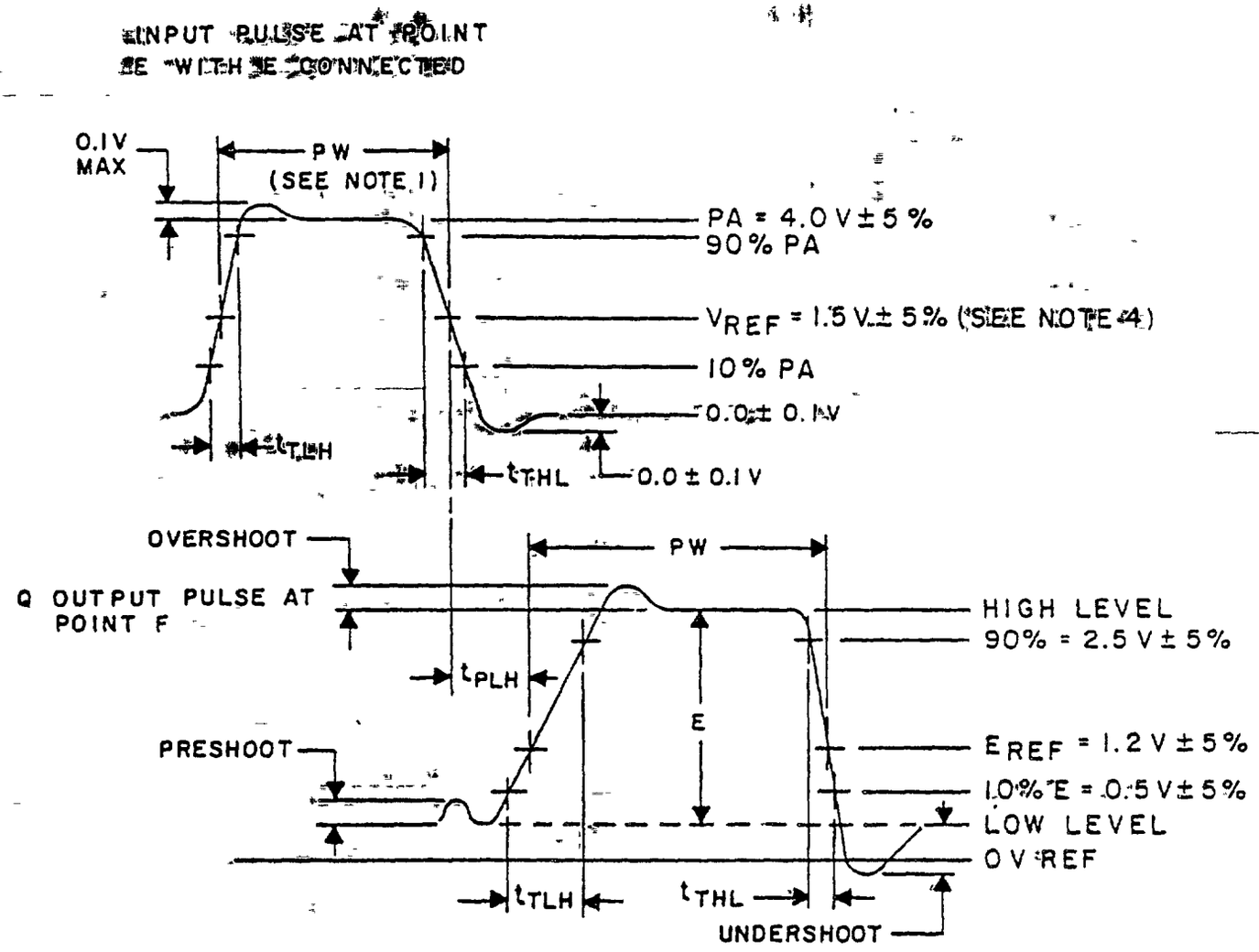
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NOTES:
1. R1 = 50 Ω ±5%, 1 W.
2. For test conditions J1-J8 use item D03 and items Y04-1 and Y04-2, measurements must have 5mV/mm resolution.
3. Adjust item P01-1 to 5.5V ±1% for test conditions F1, F4, F5, F8, F10, F11, F13, F16, F18, F19, F22, F23, F26, F27, F29, F32, F34, F35, F38, F39, F42, F43, F46, F47, F50, F51, F54, F55, F57 and F60, and 4.5V ±1% for test conditions F2, F3, F6, F7, F9, F12, F14, F15, F17, F20, F21, F24, F25, F28, F30, F31, F33, F36, F37, F40, F41, F44, F45, F48, F49, F52, F53, F56, F58, F59, G1-G8, H1-H4, J1-J8 and K1-K12.

FIGURE 8. Output pulse width (Q and $\overline{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level) and output pulse (overshoot, undershoot and preshoot) test circuit for tables VI through X.

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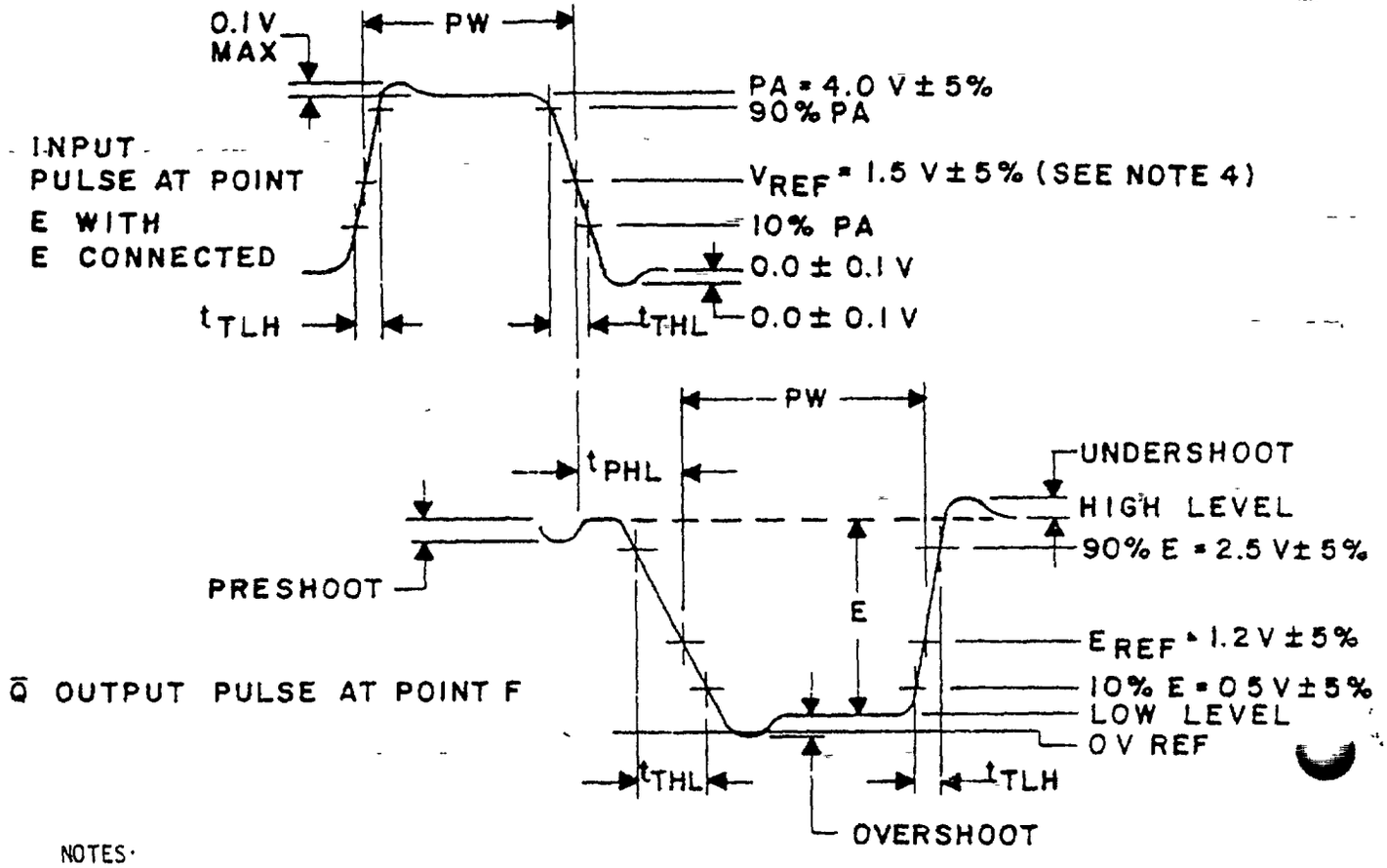


NOTES:

1. Input pulse parameters:
   PA = 4.0 V ±5%.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PW = 125 ns ±10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60 and G1-G4.
   = 100 µs ±10% for test conditions F9-F12, F17-F28 and F33-F56.
   PRP = 300 µs ±10% for test conditions F1-F8, F13-F16, F29-F32, F51-F60 and G1-G4.
   = 200 µs ±10% for test conditions F9-F12, F17-F28 and F33-F56.
2. For test conditions G1-G8, H1-H4, J1-J8 and K1-K12, connect item Y01-2 as close as physically possible to the module pin.
3. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5 V amplitude.
4. Input pulse width is measured at 50% PA for test conditions M1-M12.

FIGURE 9. Output pulse width (Q and $\bar{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII.

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NOTES:

1. Input pulse parameters:
   PA = 4.0 V ±5%.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PW = 125 ns ±10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60 and G1-G4.
   = 100 µs ±10% for test conditions F9-F12, F17-F28, and F33-F56.
   PRP = 300 µs ±10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60 and G1-G4.
   = 200 µs ±10% for test conditions F9-F12, F17-F28 and F33-F56.
2. For test conditons G1-G8, H1-H4, J1-J8 and K1-K12, connect item Y01-2 close as physically possible to the module pin.
3. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5 V amplitude.
4. Input pulse width is measured at 50% PA for test conditions M1-M12.

FIGURE 10. Output pulse width (Q and $\bar{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII.

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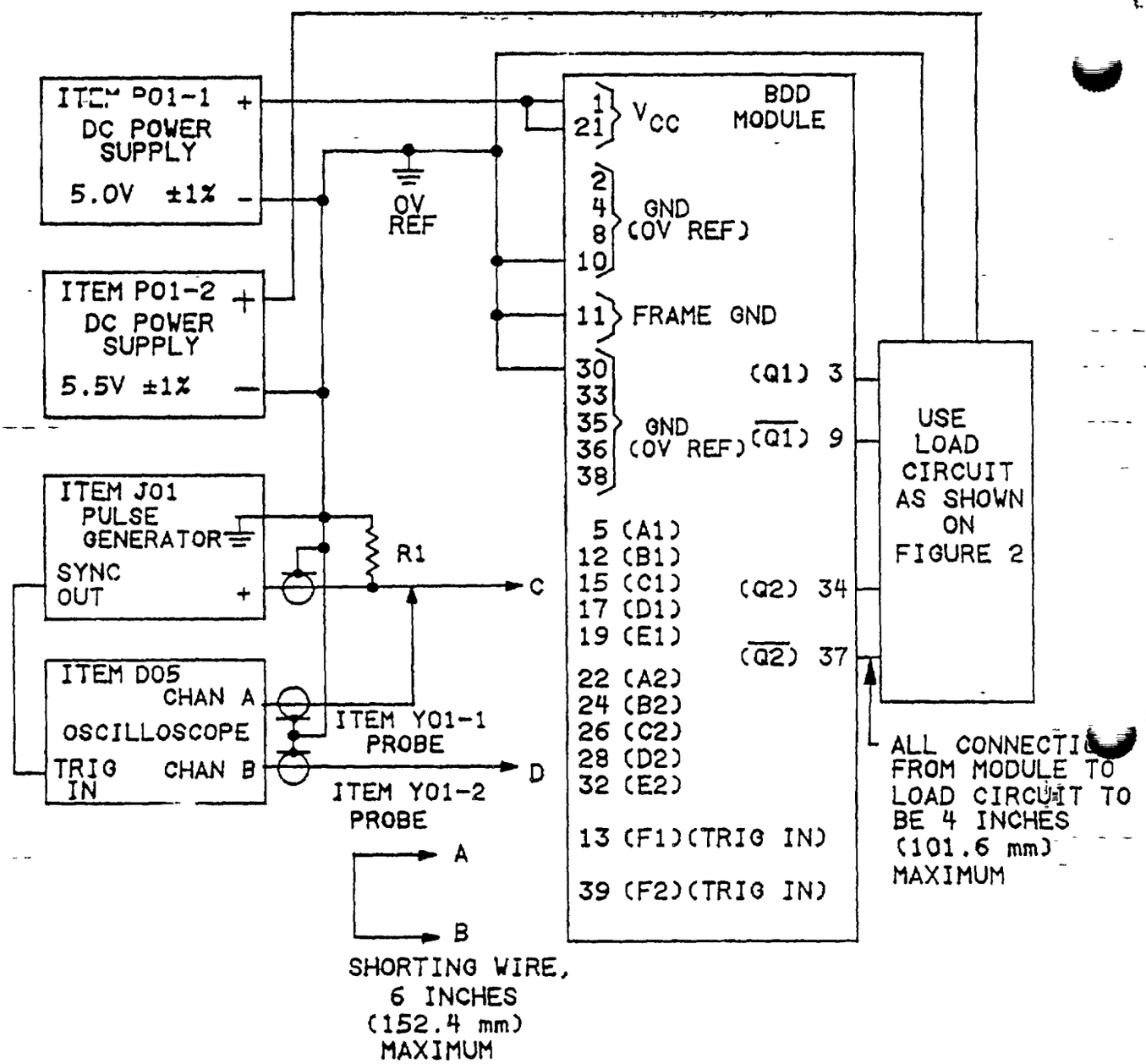
TABLE XI. Pulse width recovery time test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	Q1 Pulse width recovery time	5	19	13	3	0.09	0.12	µs	11,12
L2	Q1	Open	Open	13	3	1.39	1.61		
L3	Q2	22	26	39	34	0.09	0.12		
L4	Q2	Open	Open	39	34	1.39	1.61		

TABLE XII. Input sensitivity (trigger) and immunity test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Q1 Input sensitivity, trigger	5	19	13	3	0.09	0.12	µs	9,11,12
M2	$\overline{Q1}$	5	19	13	9				10,11,12
M3	Q2	22	26	39	34				9,11,12
M4	$\overline{Q2}$	22	26	39	37				10,11,12
M5	Q1 Input immunity, 20 ns	5	19	13	3	0	5	ns	9,11,12
M6	$\overline{Q1}$	5	19	13	9				10,11,12
M7	Q2	22	26	39	34				9,11,12
M8	$\overline{Q2}$	22	26	39	37				10,11,12
M9	Q1 Input immunity, 0.95 V	5	19	13	3				9,11,12
M10	$\overline{Q1}$	5	19	13	9				10,11,12
M11	Q2	22	26	39	34				9,11,12
M12	$\overline{Q2}$	22	26	39	37				10,11,12

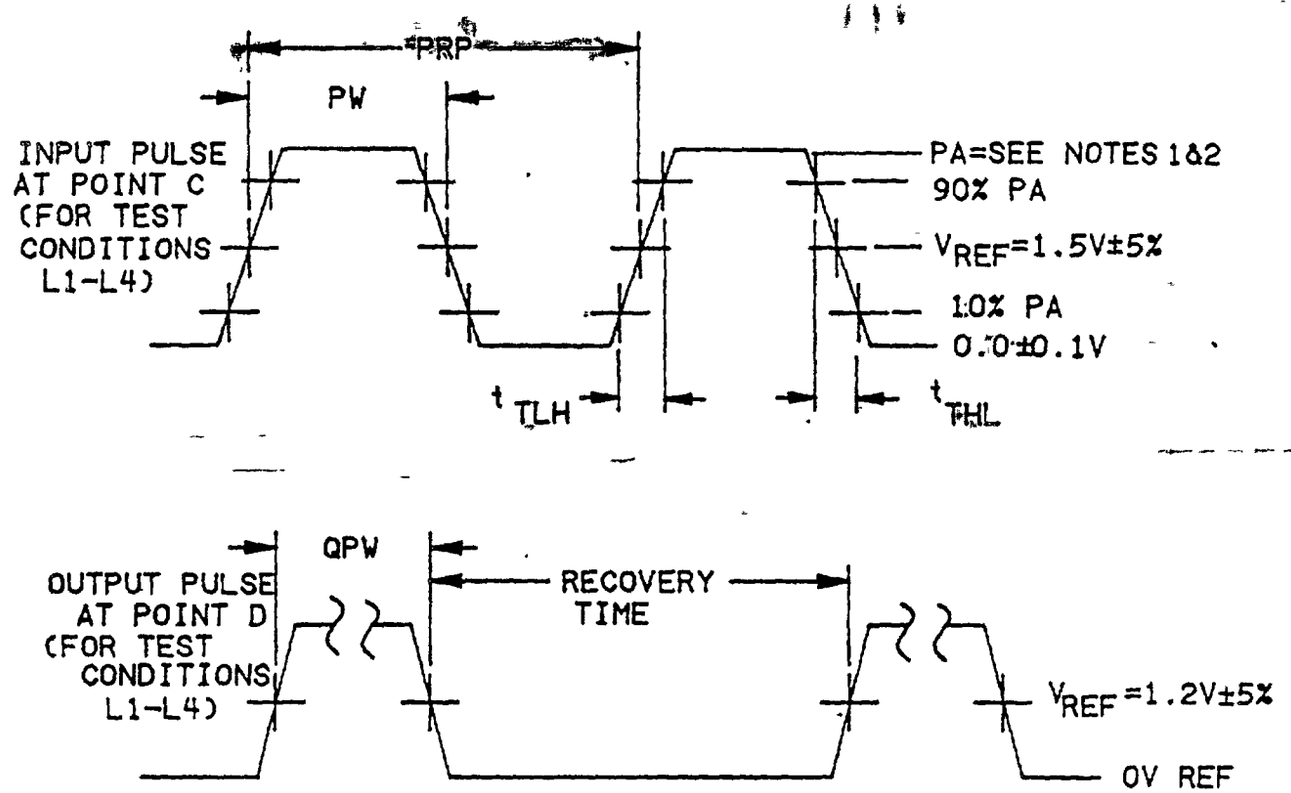
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NOTE R1 = 50 Ω ±5%, 1 W.

FIGURE 11. Pulse width recovery time, input sensitivity (trigger) and immunity test circuit for tables XI ar XII

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NOTES:
1. Input pulse parameters (test conditions L1 and L3):
   (a) PA (reference to 0V) = 4 V ±5%.
   (b) PW (1.5 V level) = 125 ns ±10%.
   (c) $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   (d) PRP = 0.6 µs ±5%.
2. Input pulse parameters (test conditions L2 and L4):
   (a) PA (reference to 0V) = 4 V ±5%.
   (b) PW (1.5 V level) = 125 ns ±10%.
   (c) $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   (d) PRP = 2.0 µs ±5%.
3. Input pulse parameters (test conditions M1 through M4):
   (a) PA (reference to 0V) = 3.2 V ±5%.
   (b) PW = 135 ns ±5%.
   (c) $t_{TLH} = t_{THL}$ = 40 ns ±5%.
   (d) PRR = 0.5 MHz ±10%.
4. Input pulse parameters (test conditions M5 through M8):
   (a) PA (reference to 0V) = 5.0 V ±5%.
   (b) PW = 20 ns ±5%.
   (c) $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   (d) PRR = 0.5 MHz ±10%.
5. Input pulse parameters (test conditions M9 through M12):
   (a) PA (reference to 0V) = 0.95 V ±5%.
   (b) PW = 1000 ns ±5%.
   (c) $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   (d) PRR = 0.5 MHz ±10%.
6. For test conditions M5 through M12, if a pulse is present at the output of the module (point D), it shall not exceed the limits of Table I.

FIGURE 12. Pulse width recovery time, input sensitivity (trigger) and immunity test circuit waveforms for tables XI and XII.

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TABLE XIII. Power supply (transient rise time and false trigger) test.

Test condition	Parameter	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	$\overline{Q1}$ Power supply transient rise time	9	9.0	53.0	µs	13
N2	$\overline{Q2}$	37	9.0	53.0		
N3	$\overline{Q1}$	9	1.5	40.0		
N4	$\overline{Q2}$	37	1.5	40.0		
N5	$\overline{Q1}$ Power supply false trigger	9	2.0	5.5	V	
N6	$\overline{Q2}$	37				
N7	$\overline{Q1}$	9				14
N8	$\overline{Q2}$	37				14

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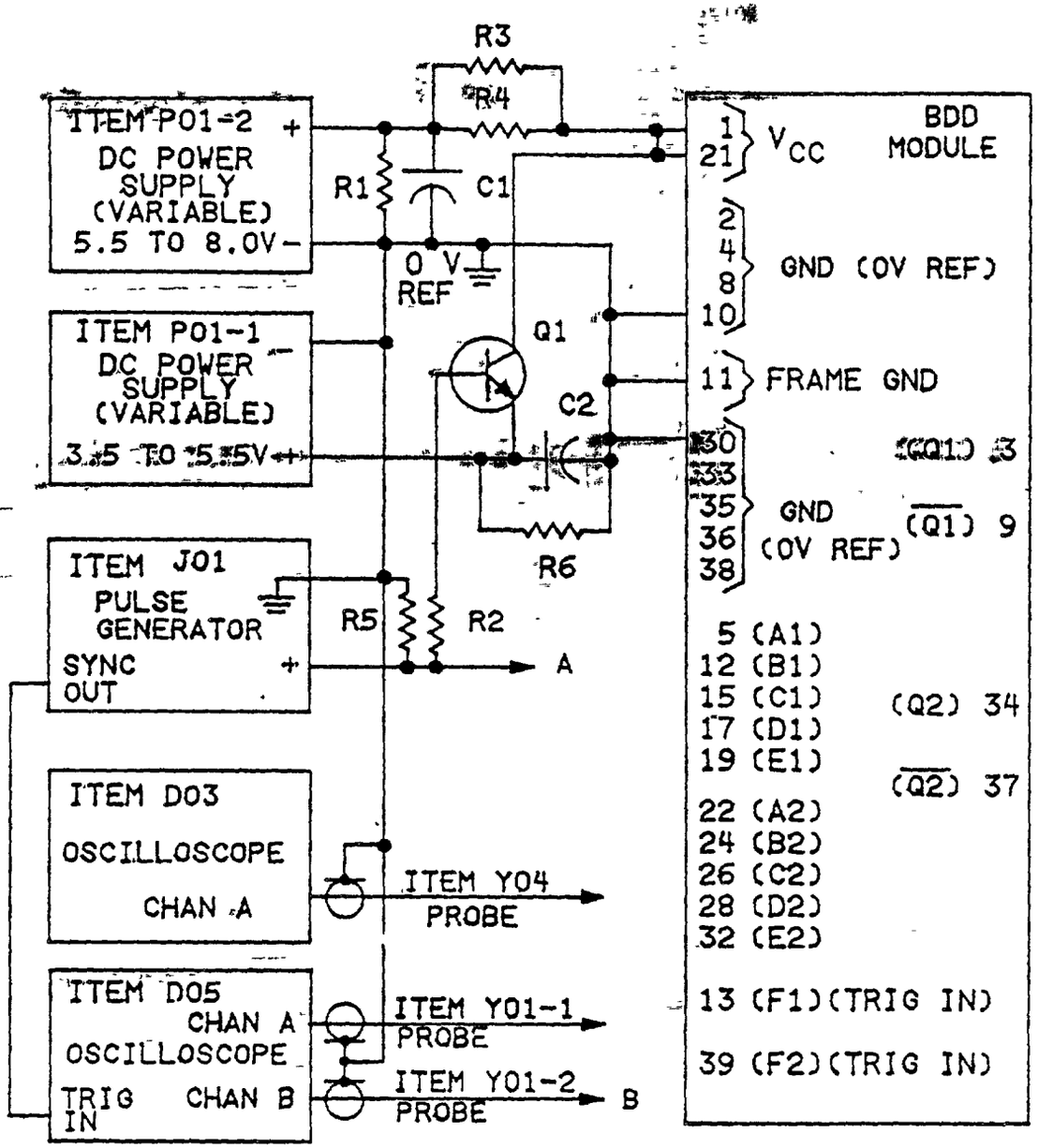
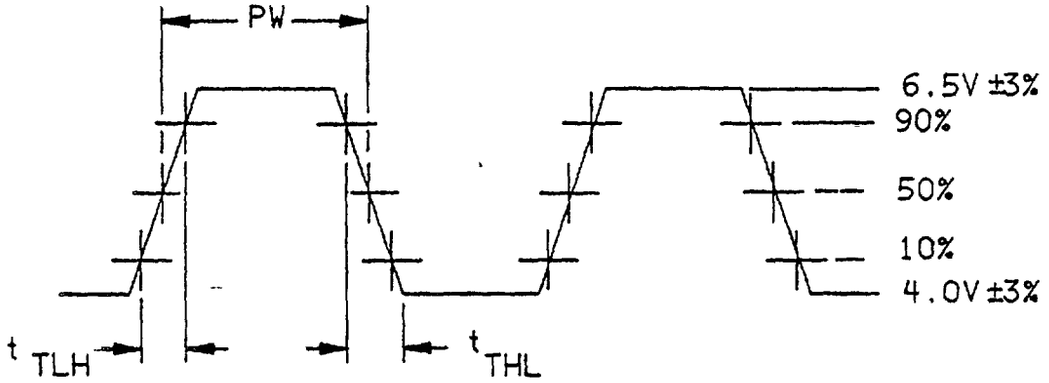


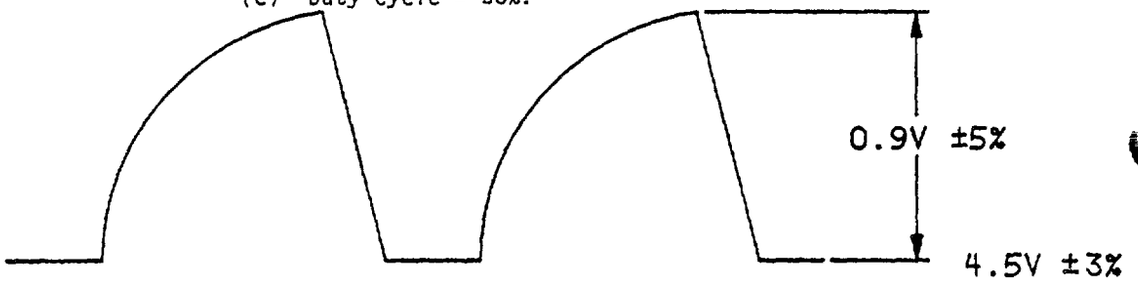
FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XIII.

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NOTES:
1. R1 = R6 = 18.2 Ω ±1%, 3 W.
2. R2 = R5 = 51.1 Ω ±1%, 1/2 W.
3. R3 = R4 = 10 Ω ±1%, 1 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N3108A or equivalent.

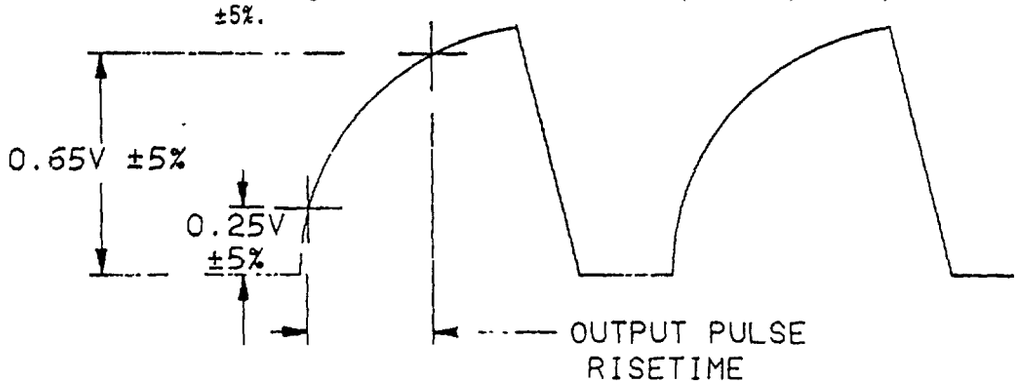


6. Using item D03, adjust item J01 to obtain the above waveform at point A with the output of item J01 disconneted from R2.
   (a) $t_{TLH} = t_{THL}$ = 50 ns ±20%.
   (b) PW = 400 µs ±10%.
   (c) Duty cycle = 50%.



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N1 through N6 connect item J01 to R2; connect item Y04 probe to module pin 1; and adjust P01-2 to set the baseline of the voltage pulse at module pin (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5%.

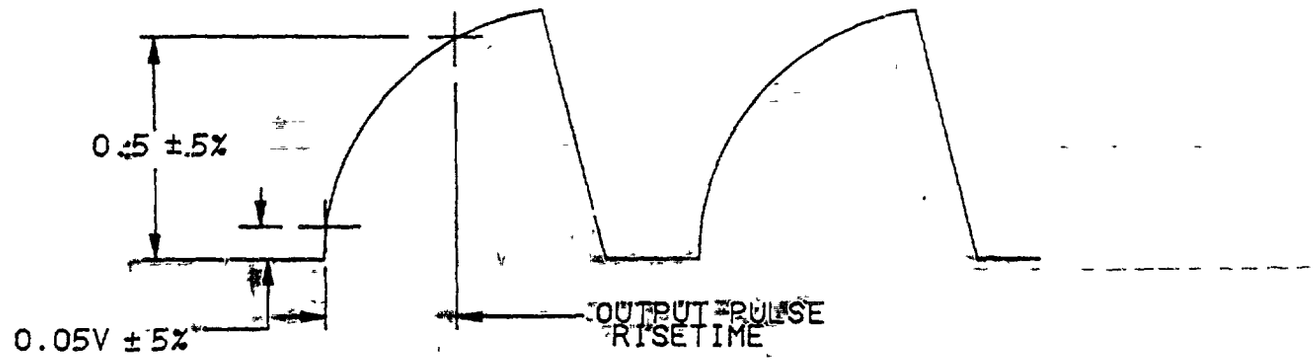


8. For test conditions N1 and N2, the output pulse rise time shall be measured as shown above and shall conform to the requirements of table I using item D05 and item Y01-2 (with 1X attenuator and AC coupler).

FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XIII (cont'd).

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9. For test conditions N3 and N4, the output pulse rise time shall be measured as shown below and shall conform to the requirements of table I using item D05 and item Y01-2 (with 1X attenuator and AC coupler).



10. For test conditions N5 and N6 the output pulse measurement (R) shall conform to the requirements of table I using item D03. The value R is defined as the minimum voltage of the waveform.

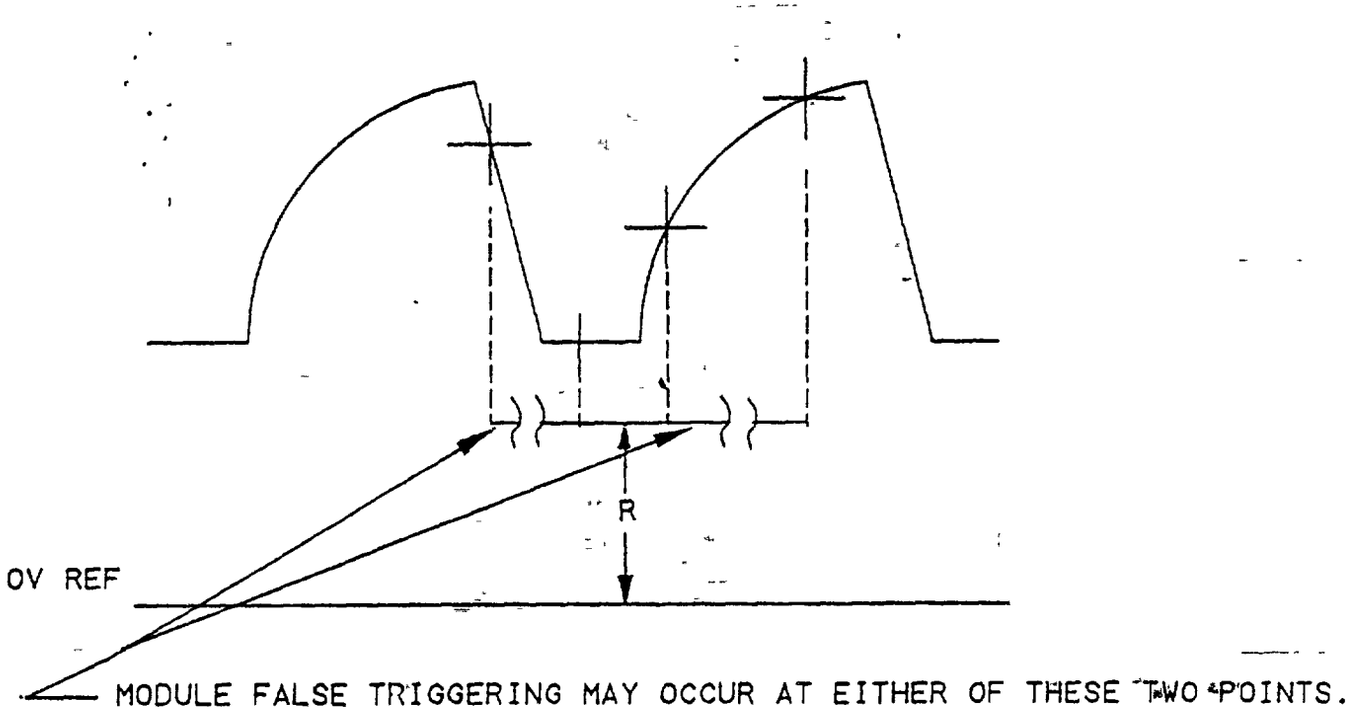


FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XIII (cont'd).

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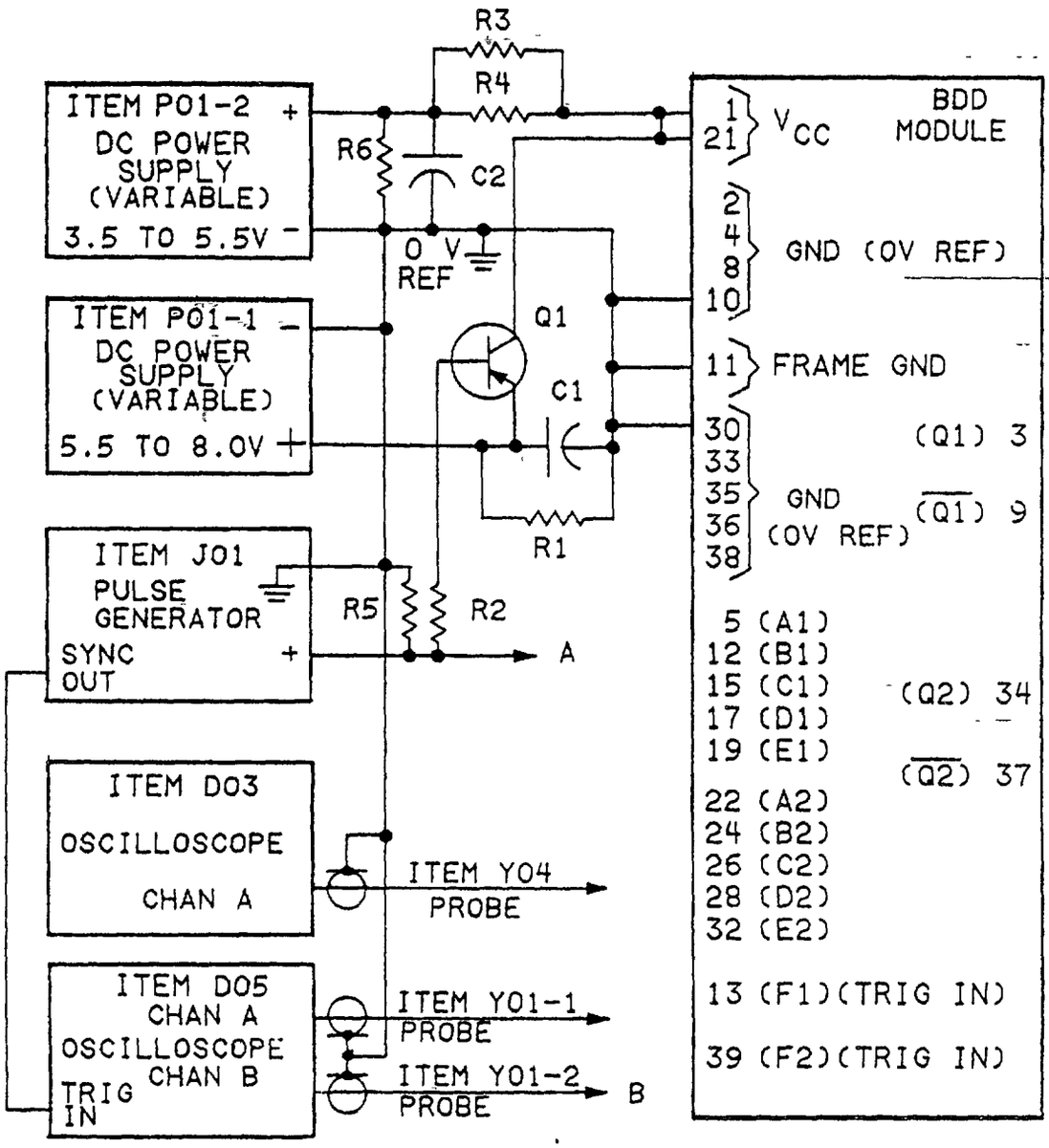
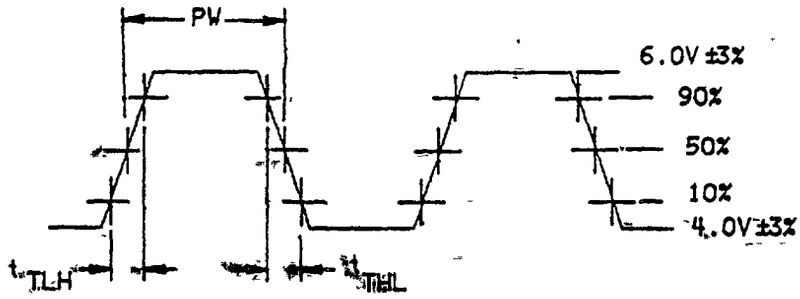


FIGURE 14. Power supply (false trigger) test circuit for table XIII.

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NOTES:
1. R1 = R6 = 18.2Ω ±1%, ¼W.
2. R2 = R5 = 51.1Ω ±1%, 1/2 W.
3. R3 = R4 = 10Ω ±1% 3/4 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N4029 or equivalent.

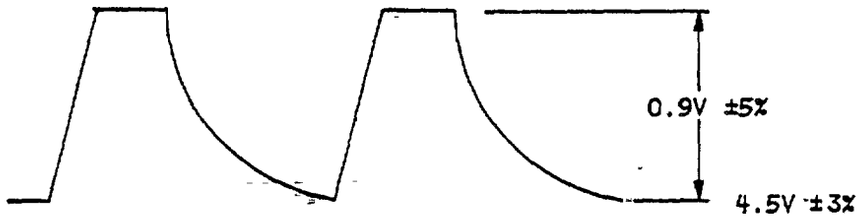


6. Using item D03, adjust item J01 to obtain the above waveform at point A with the output of item J01 disconnected from R2.

   (a) $t_{TLH} = t_{THL}$ = 50 ns ±10%

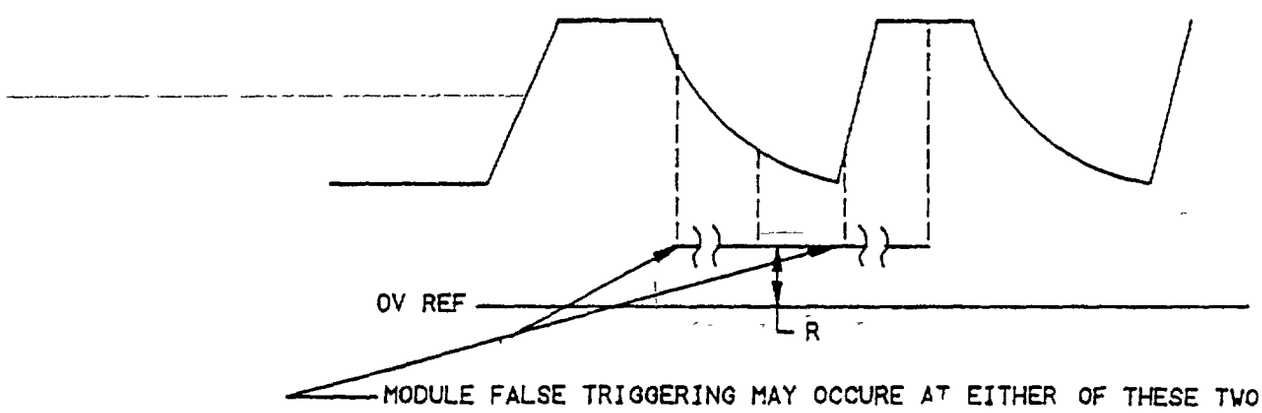
   (b) PW = 400 µs ±10%

   (c) Duty cycle = 50%



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N7 and N2 connect item J01 to R2; connect item Y04 probe to module pin 1, and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9V ±5%.



OUTPUT PULSE

8. For test conditions N7 and N8 the output pulse measurement (R) shall conform to the requirements of table I using item D03. The value R is defined as the minimum voltage of the waveform.

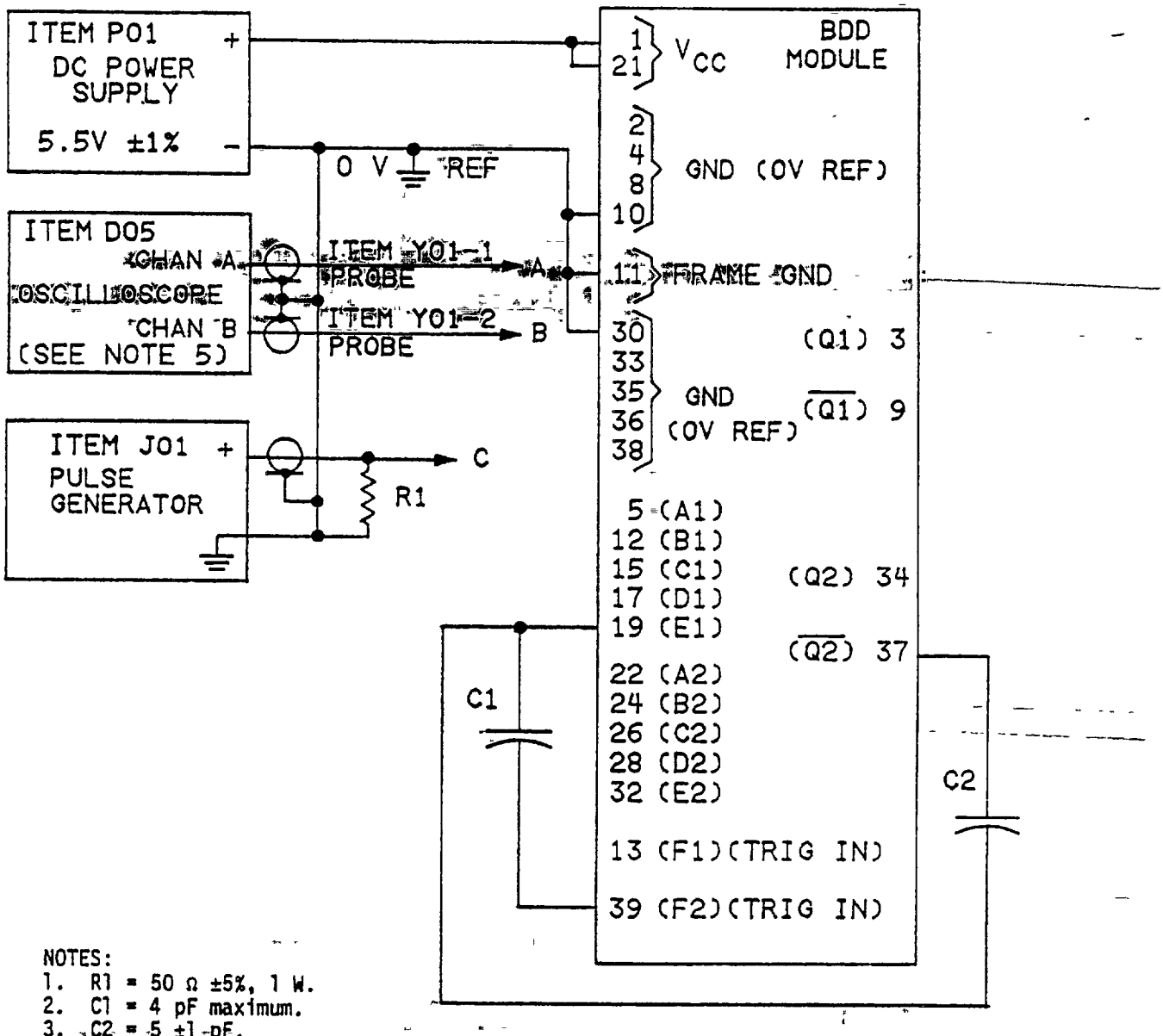
FIGURE 14. Power supply (false trigger) test circuit for table XIII (cont'd).

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TABLE XIV. Crosstalk test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q1 Crosstalk	39	3	39	0	5	ns	15,17
P2	Q1	37	3					
P3	Q1	37	3					
P4	$\overline{Q1}$	39	9					
P5	$\overline{Q1}$	37	9					
P6	$\overline{Q1}$	37	9					
P7	Q2	13	34	13				16,17
P8	Q2	9	34					
P9	Q2	9	34					
P10	$\overline{Q2}$	13	37					
P11	$\overline{Q2}$	9	37					
P12	$\overline{Q2}$	9	37					

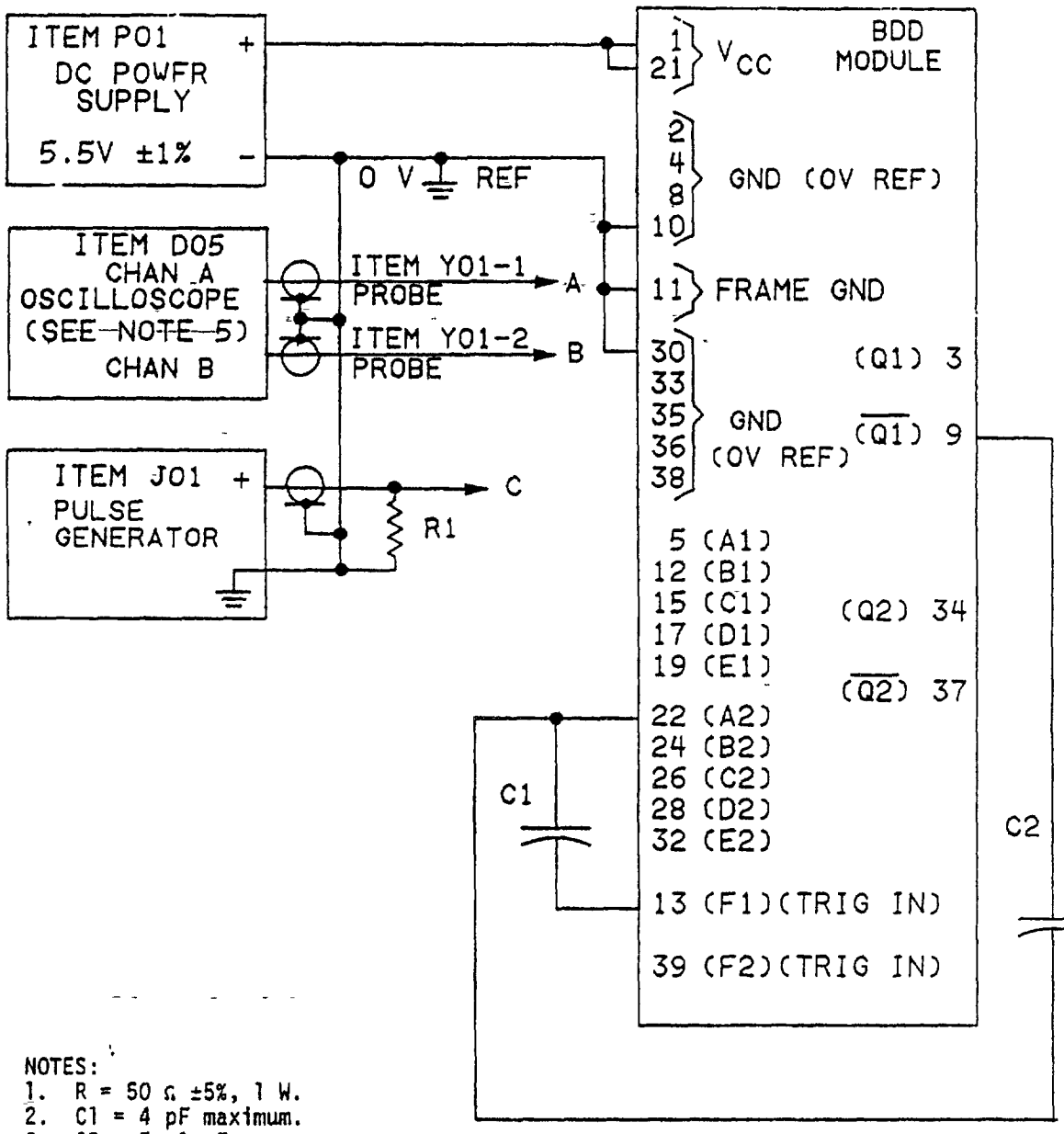
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NOTES:
1. R1 = 50 Ω ±5%, 1 W.
2. C1 = 4 pF maximum.
3. C2 = 5 ±1 pF.
4. The capacitances (C1 and C2) are measured using item C02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not item Y01 probe capacitance. Glass trimmer capacitors may be added to conform to the required values.
5. Item D05 internal trigger slope shall be negative for test conditions P1, P3, P4, and P6 and positive for test conditions P2 and P5.

FIGURE 15. Crosstalk test circuit for table XIV.

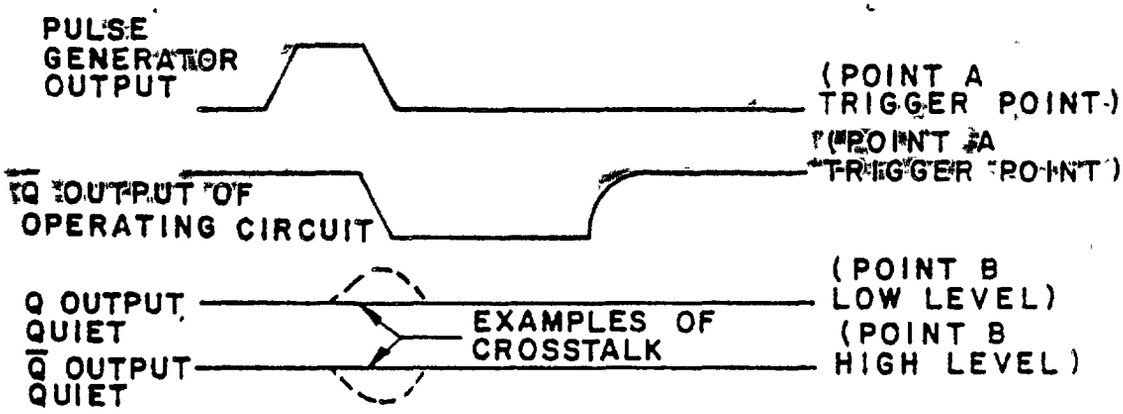
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NOTES:
1. R = 50 Ω ±5%, 1 W.
2. C1 = 4 pF maximum.
3. C2 = 5 ±1 pF.
4. The capacitances (C1 and C2) are measured using item G02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not item Y01 capacitance. Glass trimmer capacitors may be added to conform to the required values.
5. Item D05 internal trigger slope shall be negative for test conditions P7, P9, P10 and P12 and positive for test conditions P8 and P11.

FIGURE 16. Crosstalk test circuit for table XIV.

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NOTES:
1. Adjust the output of item J01 to obtain the following input pulse:
   PA = 4 V ±5%.
   PW (2 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ns ±10%.
   PRP = 3 µs ±10% (use items D03 and Y04 for this measurement).
2. To examine the output waveforms under test, item D05 shall be internally triggered using the signal on channel H (item Y01-1).
3. For each test condition, the maximum pulse width detected at point B, measured at 1.0 V above the low level on the Q output and at 2.0 V below the high level on the Q̄ output, shall not excees the limits of table I.

FIGURE 17. Crosstalk test circuit notes and waveforms for table XIV.

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TABLE XV. Zener applied potential and trigger voltage high test.

Test condition	Parameter	Switch		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		S1	S2	A to pin	Min	Max		
Q1	$\overline{Q1}$ Zener applied potential	Open	Closed	9	3.8	5.5	V	18
Q2	$\overline{Q2}$ Zener applied potential	Open	Closed	37	3.8	5.5		
Q3	Q1 Trigger voltage high	Closed	Open	3	0.0	0.4		
Q4	Q2 Trigger voltage high	Closed	Open	34	0.0	0.4		

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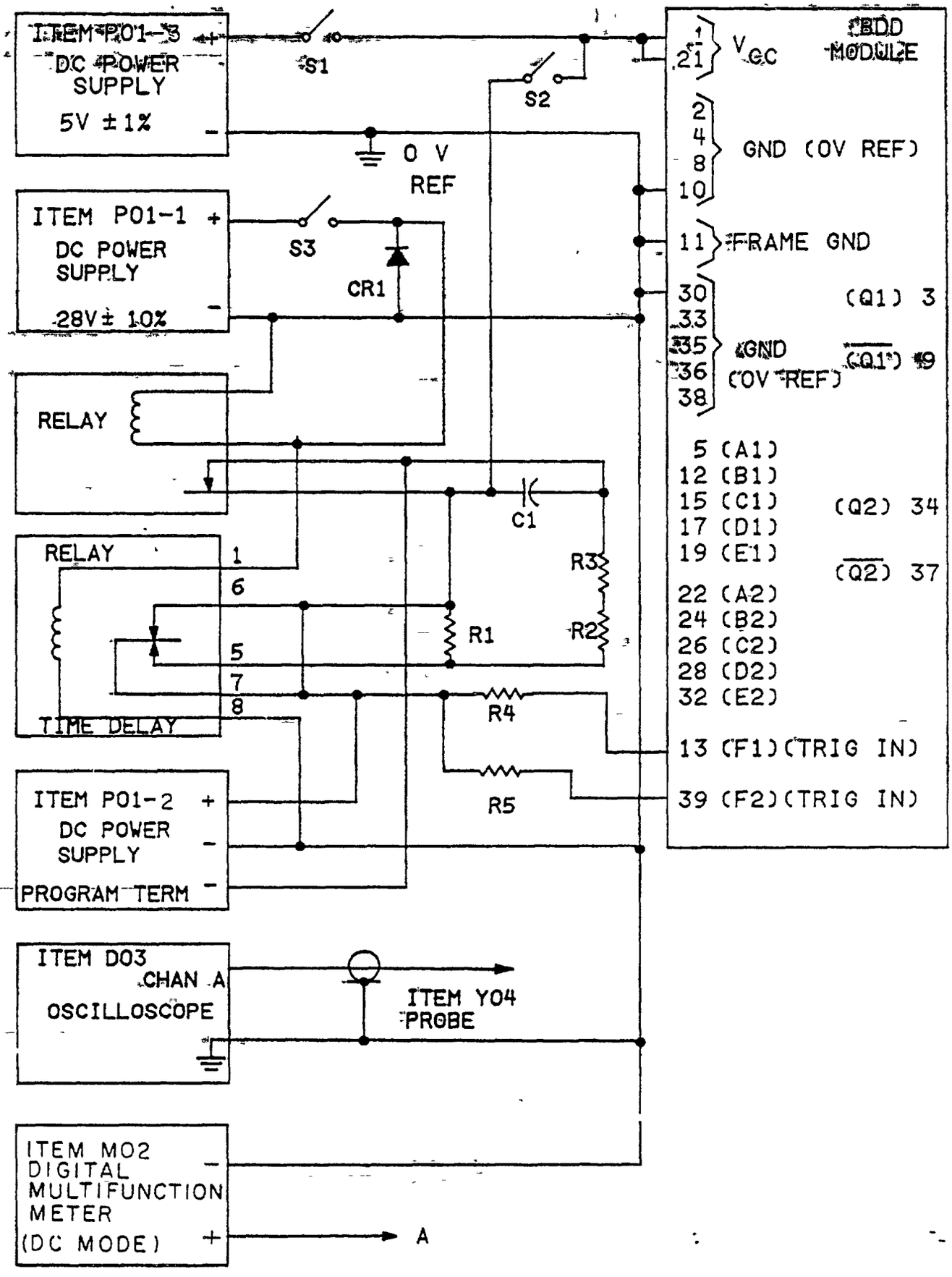
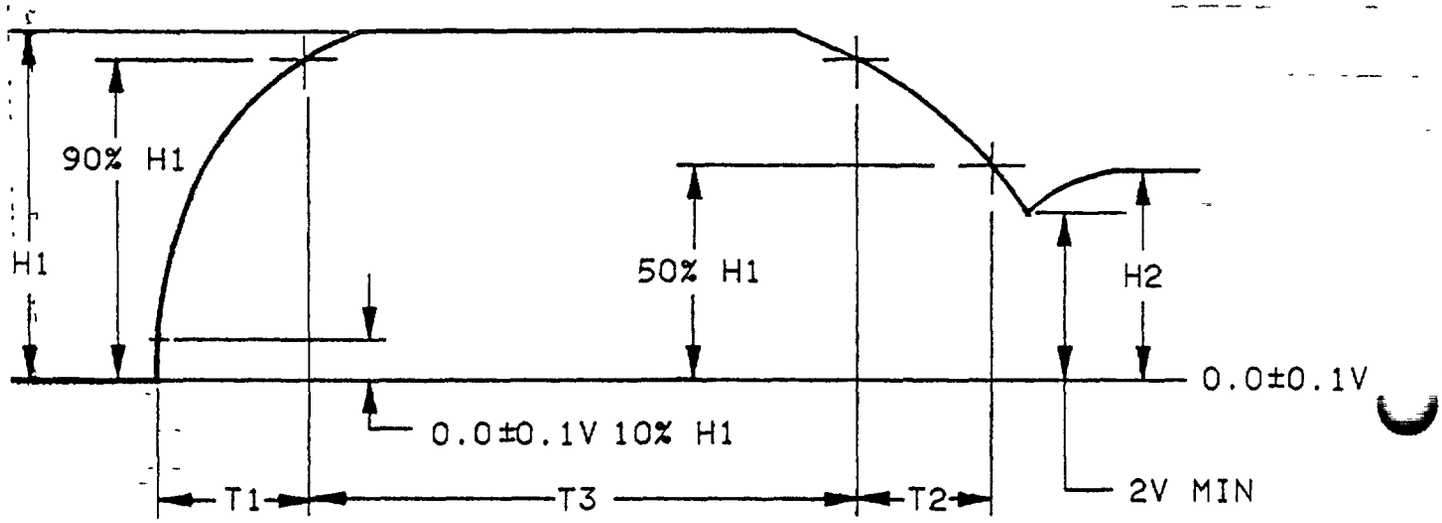


FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XV.

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NOTES:

1. R1 = 649Ω ±1%, 1/8 W.
2. R2 = 301Ω ±1%, 1/8 W.
3. R3 = 178Ω ±1%, 1/8 W.
4. R4 = R5 = 154Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647 or equivalent.
7. Relay 1 = 28V relay.
8. Relay 2 = Agastat model 2112 or equivalent.
9. Resistors R3, R4, R5, and capacitor C1 are programming components for item P01-2 and are based on a programming constant of 100Ω per volt.
10. Using item D03 with item Y04 probe, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay after switch S3 is closed.



Waveform parameters:

(a) H1 = 11.25V ±1%
(b) H2 = 5.0V ±10%
(c) T1 = 100 $^{+100}_{-50}$ ms
(d) T2 = 100 $^{+400}_{-50}$ ms
(e) T3 = 0.5 sec ±10%

11. Item P01-2 should be selected to ensure that the programming terminal is floating (-).
12. Test sequence: (a) Close switch S3, (b) wait 2 seconds minimum, (c) make voltage measurements for test conditions, and (d) open switch S3.

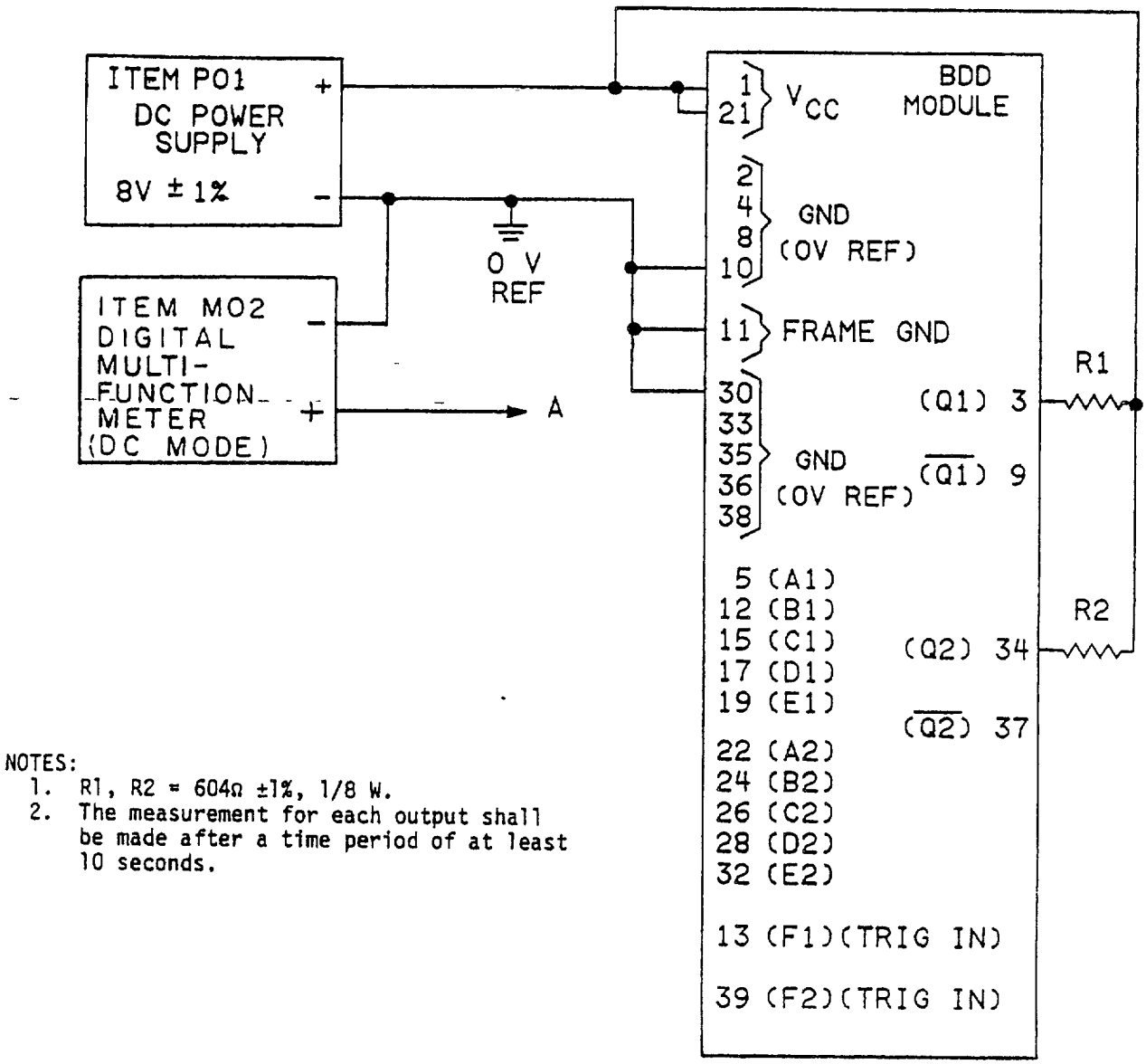
FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XV (cont'd).

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TABLE XVI. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
R1	Q1 V_{OUT} (low loaded)	3	0.0	1.1	V	19
R2	Q2 V_{OUT} (low loaded)	34	0.0	1.1	V	19

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NOTES:
1. R1, R2 = 604Ω ±1%, 1/8 W.
2. The measurement for each output shall be made after a time period of at least 10 seconds.

FIGURE 19. V_{OUT} (low-loaded) test circuit for table XVI.

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